

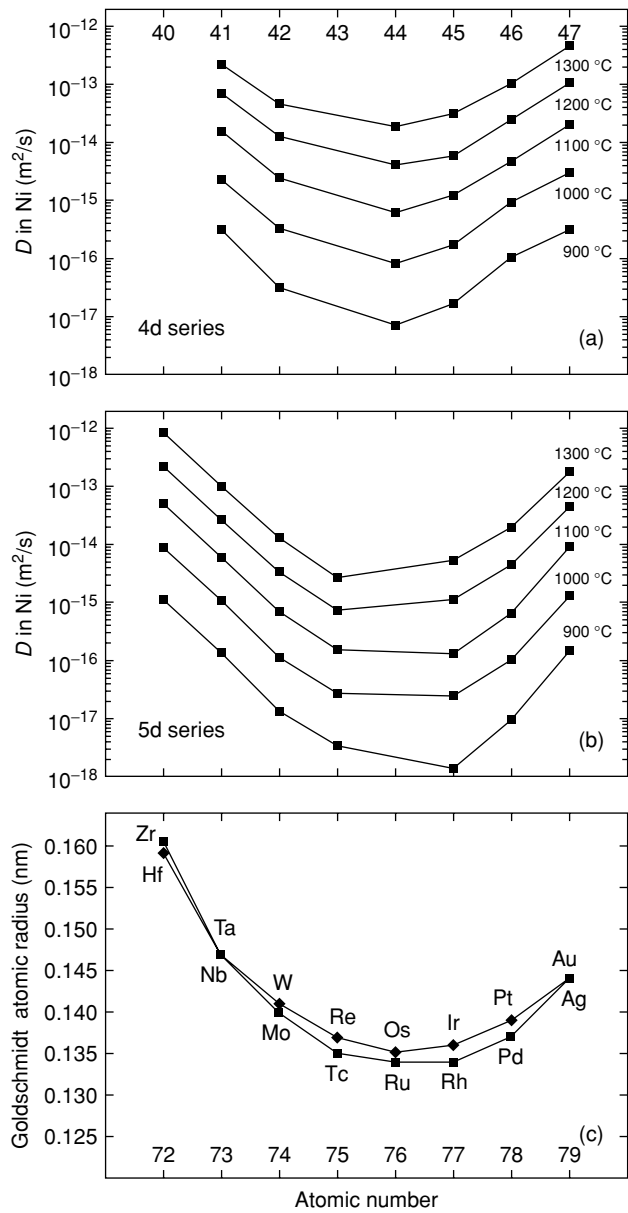


Fig. 2.34. Measurements of the interdiffusion coefficient in binary Ni/Ni-X diffusion couples: (a) for the 4d series of the d block of transition metals, (b) for the 5d series, and (c) variation of the Goldschmidt radius of the transition metals with atomic number [29,30].

experimentally [29,30], by analysing diffusion couples fabricated from Ni and Ni-X alloys, where X is a transition metal. Elements near the centre of the d block of the periodic table, for example, Re and Ru, were found to diffuse several orders of magnitude more slowly than those from the far west or far east of the d block, for example, Hf and Au; see Figure 2.34. These experimental results are interesting, since they disprove the widely held

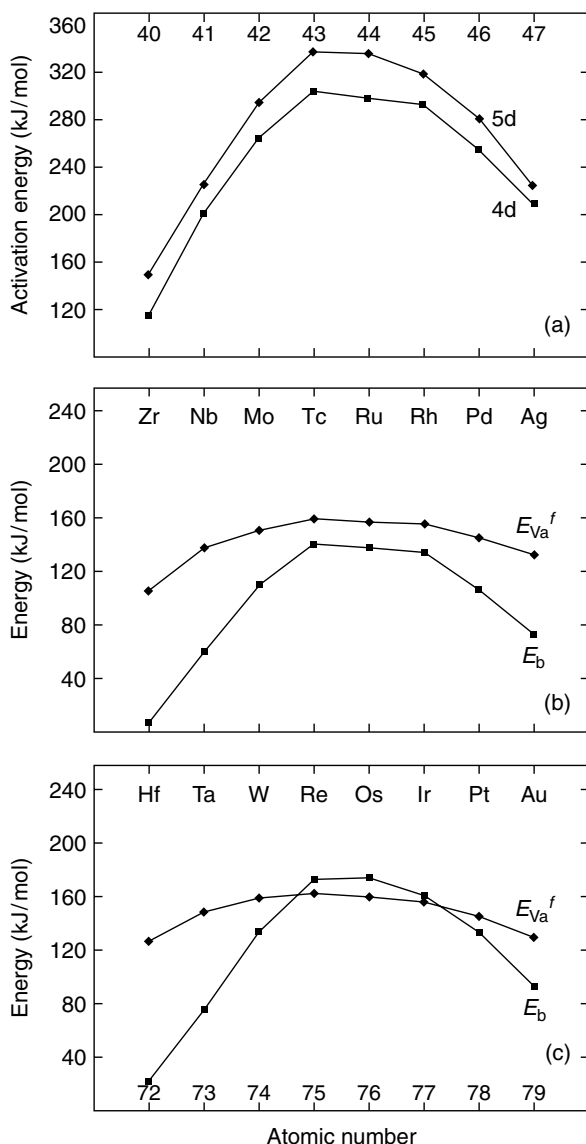


Fig. 2.35. Results of calculations from a model of the diffusion behaviour of the 4d and 5d transition metals in nickel [31,32]: (a) the activation energy for diffusion, and the partition of it into vacancy and barrier energy terms for the 4d and 5d solutes in (b) and (c), respectively.

view that diffusion is least rapid when the lattice misfit between solute and solvent atom radii is the greatest. These observations have been rationalised using quantum-mechanical modelling using first principles methods [31,32]; this has allowed an evaluation of the activation energies for solute–vacancy exchange, denoted E_b , and the vacancy formation energy for a solute–vacancy complex, denoted $E_{X-v_a}^f$. The calculations indicate (see Figure 2.35) that $E_{X-v_a}^f$ does not vary strongly as one crosses the d block and by no more than 40 kJ/mol; in

fact, the magnitudes of $E_{\text{Hf-Va}}^f$ and $E_{\text{Au-Va}}^f$ are close to 120 kJ/mol, and thus are close to the vacancy formation energy in pure Ni, implying that the solute–vacancy binding energy for elements such as Hf and Au is negligible. The Re–Va binding energy is about +40 kJ/mol. Thus, the dependence of $E_{\text{X-Va}}^f$ on atomic number is relatively weak. Instead, the major contribution to the variation of the interdiffusion coefficient with atomic number arises from the diffusion energy barrier, E_b . This is significant for elements such as Re and Ru. Alloying with these elements, which are amongst the densest in the d block owing to their electronic bonding and many unpaired electrons, causes directional and incompressible Ni–Re and Ni–Ru bonds which do not favour solute–vacancy exchanges. This effect dominates any influences of atomic radius, misfit strain and differences in the solute–vacancy binding energy. It is probable that this is the reason why Re has such a significant effect on the high-temperature properties of the superalloys.

2.2.2 Defects in the gamma prime phase

A Planar defects – the anti-phase boundary

An important planar defect found in the nickel alloys is the *anti-phase boundary*, the meaning of which can be understood in the following way. Consider two perfect crystals of γ' displaced by $a/2\langle 1\bar{1}0 \rangle$; this vector links neighbouring Ni and Al atoms, but note that it is not a lattice vector. When bonded together, an interface known as the anti-phase boundary (APB) separates the otherwise perfect lattices; in the vicinity of the APB, the number of Ni–Al bonds is substantially reduced since ‘forbidden’ Ni–Ni and Al–Al bonds are formed there. The net effect is to cause an APB energy which is substantial – usually of order 100 mJ/m². The number of forbidden bonds is found to depend upon the crystallographic plane on which the APB resides [33] (see Figure 2.36); consequently, for this reason, the APB energy is anisotropic. This can be rationalised as follows. First, consider two {111} planes of the L1₂ structure stacked perfectly (Figure 2.37) – note that there are no instances in which two Al atoms are nearest neighbours. Upon displacing the second layer by $a/2\langle 1\bar{1}0 \rangle$, forbidden bonds are created: one is formed for each area, $a\sqrt{2}/2 \times a\sqrt{6}/2$ or $a^2\sqrt{3}/2$. The APB energy on the {111} plane is then $2\Omega/(a^2\sqrt{3})$, where Ω is the ordering energy, defined by $\Omega = \Omega_{\text{NiAl}} - \frac{1}{2}(\Omega_{\text{NiNi}} + \Omega_{\text{AlAl}})$, where Ω_{NiAl} , Ω_{AlAl} and Ω_{NiNi} are the Ni–Al, Al–Al and Ni–Ni bond energies, respectively. Consider next the {110} plane (Figure 2.38). One forbidden bond is formed per area $a \times a\sqrt{2}/2$ or $a^2\sqrt{2}/2$ so that the APB energy is $2\Omega/(a^2\sqrt{2})$, which is about 23% higher than for an APB on the {111} plane. Finally, consider the APB energy on the {001} plane. In this case, Figure 2.36 confirms that a translation of $a/2\langle 1\bar{1}0 \rangle$ does not cause forbidden bonds; one way to realise this is to consider the {200} planes, every second one of which consists solely of Ni atoms. It follows then that Al atoms are never brought into nearest-neighbour positions, so that the APB energy will be close to zero. In practice, the APB energy on the {001} plane is non-zero; it should be remembered that here only first-order nearest-neighbour bonding has been accounted for in these arguments.

The planar fault known as the APB is of profound importance in superalloy metallurgy. This is because the γ' phase deforms very differently from the disordered γ matrix in which

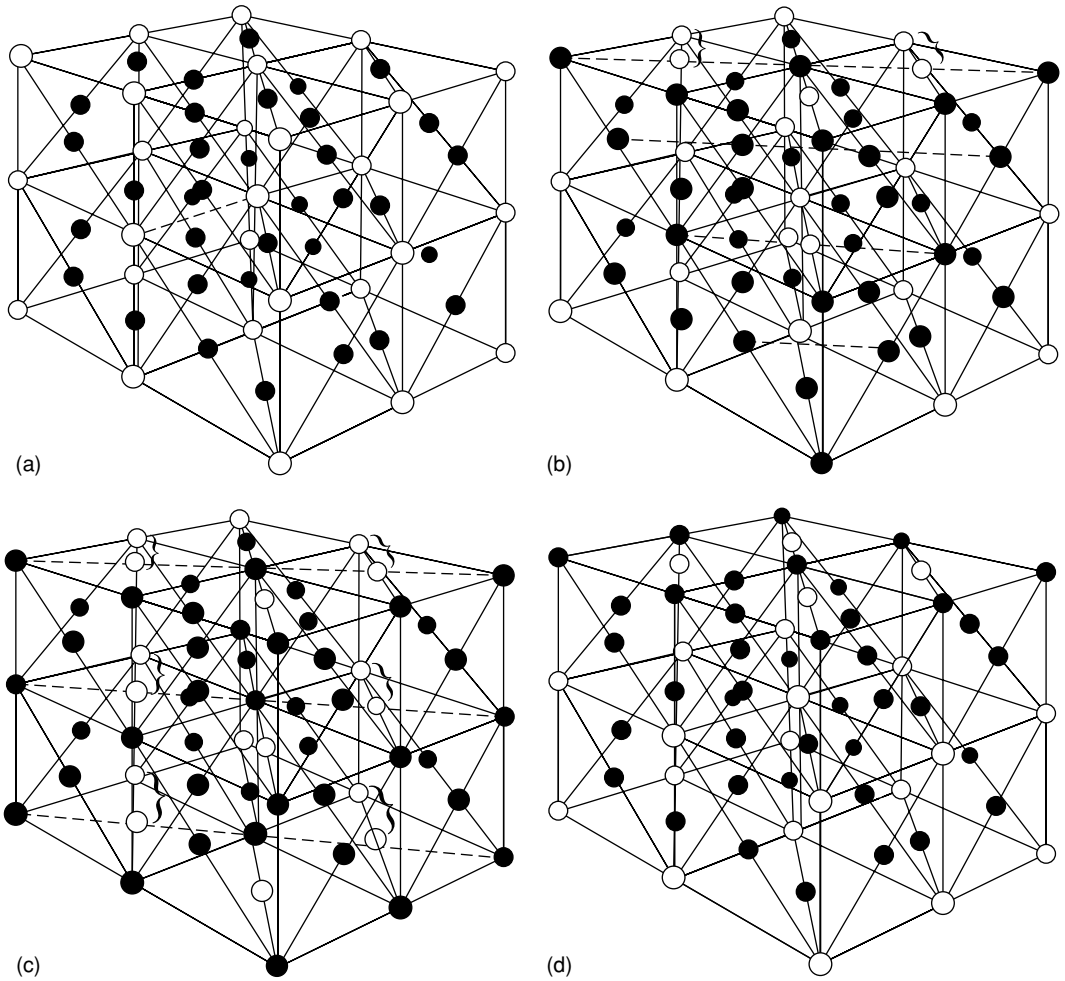


Fig. 2.36. Structure of the perfect $L1_2$ lattice, and anti-phase boundary (APB) structures caused by the APB fault lying on different planes. (a) The ordered $L1_2$ structure; (b) $(111)[110]$ anti-phase boundary in the $L1_2$ structure; (c) $(110)[110]$ anti-phase boundary in the $L1_2$ structure; (d) $(001)[110]$ anti-phase boundary in the $L1_2$ structure. Adapted from ref. [33].

it resides. In γ , slip deformation is due to dislocation glide on the $a/2\langle\bar{1}\bar{1}0\rangle\{111\}$ system; the $a/2\langle\bar{1}\bar{1}0\rangle$ lattice vector is the shortest one available, and it lies in the close-packed plane. However, in the $L1_2$ structure, the close-packed planes are again $\{111\}$, but the shortest lattice vectors, $a\langle 100\rangle$, do not reside in them. Consequently, at first sight it is not clear whether the preferred slip system should be $a\langle 100\rangle\{001\}$, $a/2\langle\bar{1}\bar{1}0\rangle\{001\}$ or $a\langle\bar{1}\bar{1}0\rangle\{111\}$; in fact, at low temperatures it is the last one which is observed. This causes the following effect. A single $a/2\langle\bar{1}\bar{1}0\rangle\{111\}$ dislocation in γ , although a perfect dislocation in that phase, cannot enter γ' without an APB fault being formed and therefore without a substantial energy penalty – this is because the closure vector required to restore the γ' lattice to its perfect state is *twice* the Burgers vector of a single dislocation in γ . It follows that $a/2\langle\bar{1}\bar{1}0\rangle\{111\}$ dislocations

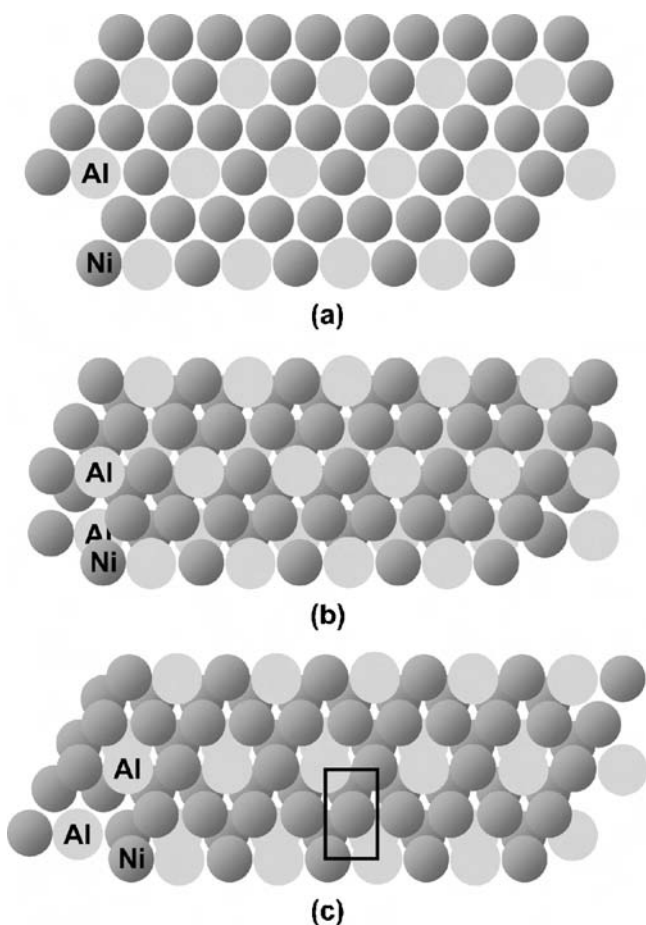


Fig. 2.37. (a) Representation of the $\{111\}$ plane of the L_{12} crystal structure of Ni_3Al . (b) Two such planes stacked according to the crystal structure. (c) Situation after displacing the second plane by $a/2(1\bar{1}0)$ on $\{111\}$. One forbidden bond is formed in the area marked.

must travel in pairs through γ' . Each dislocation is referred to as a *superpartial*, and a pair of two such dislocations is called a *superdislocation*. Since each superpartial is an imperfect dislocation in γ' , two superpartials are linked by a faulted strip, or *anti-phase boundary*. Since the associated APB energy is relatively high, and particularly when the γ' fraction is significant, the superpartials usually reside in the same γ' precipitate and are then said to be 'strongly coupled'. When the γ' fraction is smaller, so-called 'weak coupling' becomes possible. The consequences of 'strong' and 'weak' coupling of superpartial dislocations are discussed in more detail in Section 2.3.

Three other sorts of planar defect have been identified in Ni_3Al ; see, for example, the pioneering papers of Kear and co-workers [34,35]. It is helpful to consider the stacking of successive $\{111\}$ planes (see Figure 2.39) [36]; three is sufficient since this corresponds to the periodicity along $\langle 111 \rangle$, although the vector displacement of one layer above another is now twice that for the disordered γ phase. The $\{111\}$ planes are the appropriate ones to

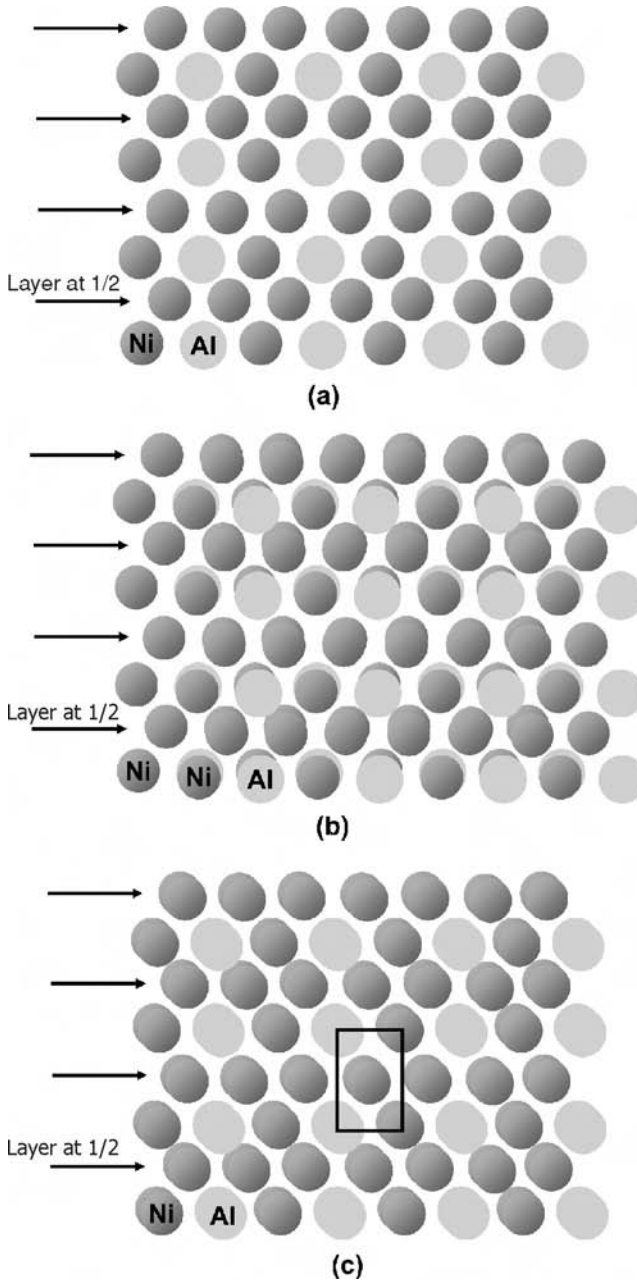


Fig. 2.38. (a) Representation of the $\{110\}$ plane of the L_{12} crystal structure of Ni_3Al . (b) Two such planes stacked according to the crystal structure. (c) Situation after displacing the second plane by $a/2(1\bar{1}0)$ on $\{110\}$. One forbidden bond is formed in the area marked.

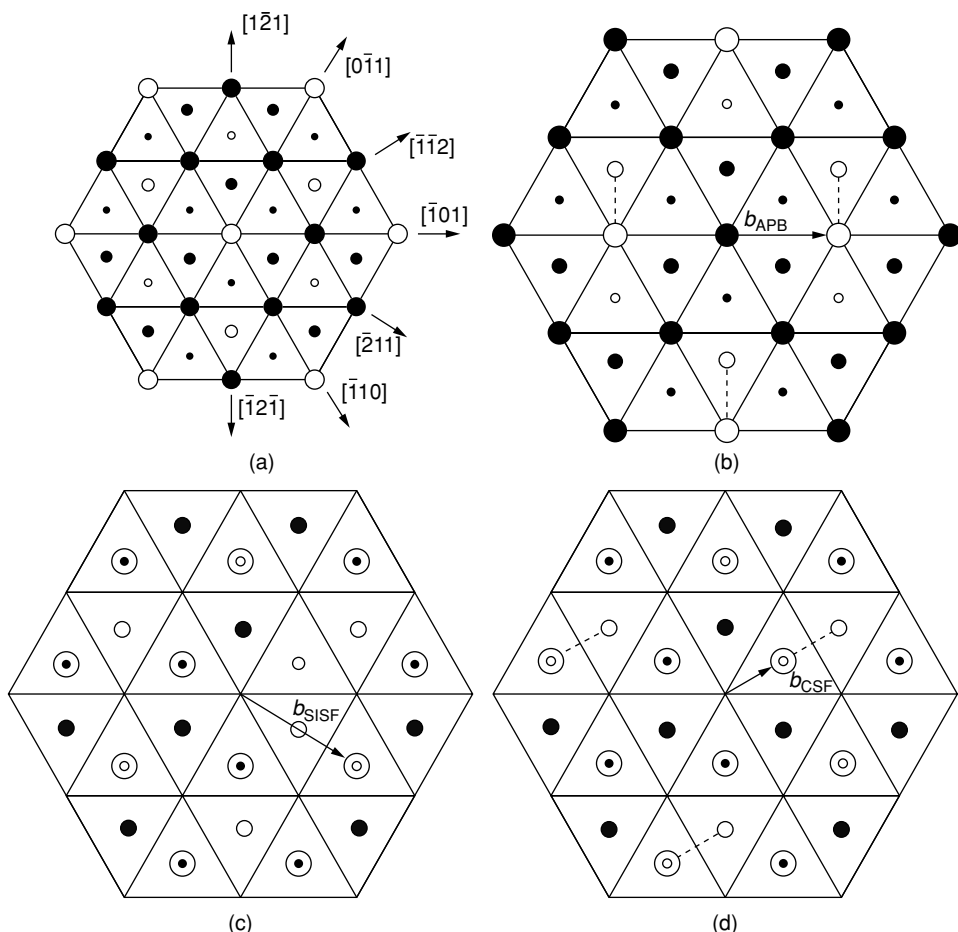


Fig. 2.39. Illustration of various faults possible in the L_{12} structure formed by shearing $\{111\}$ planes past one another: (a) three successive $\{111\}$ planes, unfaulted; (b) after shearing the top layer by $a/2\langle\bar{1}01\rangle$ to form an anti-phase boundary (APB); (c) after shearing by $a/3\langle\bar{2}11\rangle$ to form a superlattice intrinsic stacking fault (SISF); (d) after shearing by $a/6\langle\bar{1}12\rangle$ to form a complex stacking fault (CSF). Note that the large, medium and small circles represent atoms in the top, middle and bottom planes, respectively [36].

consider since planar faults in Ni_3Al arise only as a consequence of mechanical deformation; thermal treatment is insufficient – there have been no recorded observations of their occurrence due to recrystallisation, for example. If the top layer is sheared by $a/2\langle\bar{1}01\rangle$, as shown in Figure 2.39(b), one has an APB on $\{111\}$ as discussed above, and there are changes in the nearest-neighbour bonds; for example, Al–Al bonds are formed. But other displacement vectors are possible. If one shears the top layer by $a/3\langle\bar{2}11\rangle$, see Figure 2.39(c), the Al atoms in the top layer now lie directly above those in the bottom one. The stacking sequence changes from ABCABCABCABC... to ABCACABCABC..., which is equivalent to a single $\{111\}$ being removed; hence, by analogy with the intrinsic stacking fault in γ , this is referred to as a *superlattice intrinsic stacking fault* (SISF). Since it produces no

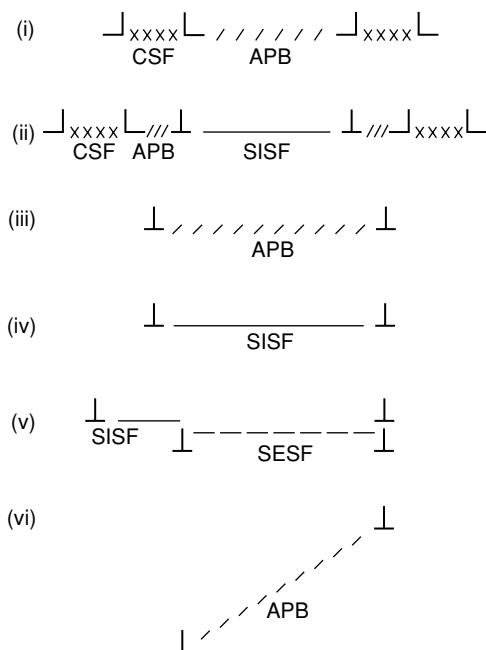


Fig. 2.40. Illustration of some possible dislocation dissociation reactions in the $L1_2$ structure. Adapted from Pope and Ezz [36].

nearest-neighbour violations, its energy is small; the energy is of order of 10 mJ/m^2 , although it is rather difficult to measure accurately. Alternatively, the top layer can be shifted by $a/6\langle\bar{1}12\rangle$ – the stacking fault vector of the FCC structure – as shown in Figure 2.39(d), so that the Al atoms in the top layer lie above directly above the Ni atoms in the bottom layer; the so-called *complex stacking fault* (CSF) then arises, which produces a change in nearest-neighbour configurations. Consequently, the CSF energy is large; for binary Ni_3Al it has been estimated [37] to be $\sim 250 \text{ mJ/m}^2$, varying by perhaps 50 mJ/m^2 across the range of stoichiometry exhibited by the compound. Finally, the *superlattice extrinsic stacking fault* (SESF) – not shown in Figure 2.39 – is produced by displacing the top layer by $a/3\langle\bar{2}11\rangle$. In this case, the shearing can be shown to be equivalent to the introduction of an extra $\{111\}$ layer. As with the SISF fault, no nearest-neighbour violations are produced, so the fault energy is low.

B Line defects – dislocations

Dislocations in the $L1_2$ structure of the γ' phase dissociate into partial dislocations, just as for a $a/2\langle\bar{1}10\rangle\{111\}$ dislocation in γ . However, the Burgers vectors of dislocations in γ' are expected to be longer than in γ as a consequence of the ordering reaction; possible closure vectors are of length $a\sqrt{2}$ or $a\sqrt{6}$, as can be seen by analysis of Figure 2.39. Hence, there is greater scope for the dissociations to be complex. The following possibilities are suggested in the review by Pope and Ezz [36], as depicted in Figure 2.40:

- (i) dissociation by CSF, APB and CSF on $\{111\}$ planes, according to

$$a[\bar{1}01] \rightarrow \frac{a}{6}[\bar{1}\bar{1}2] + \frac{a}{6}[\bar{2}11] + \frac{a}{6}[\bar{2}11] + \frac{a}{6}[\bar{1}\bar{1}2] \quad (2.17)$$

- (ii) dissociation by APB, CSF and SISF on $\{111\}$ planes, according to

$$a[\bar{1}01] \rightarrow \frac{a}{6}[\bar{1}\bar{1}2] + \frac{a}{6}[\bar{2}11] + \frac{a}{6}[\bar{1}2\bar{1}] + \frac{a}{6}[\bar{1}2\bar{1}] + \frac{a}{6}[\bar{1}\bar{1}2] + \frac{a}{6}[\bar{2}11] \quad (2.18)$$

- (iii) dissociation by APB on $\{111\}$ planes, according to

$$a[\bar{1}01] \rightarrow \frac{a}{2}[\bar{1}01] + \frac{a}{2}[\bar{1}01] \quad (2.19)$$

- (iv) dissociation by SISF on $\{111\}$ planes, according to

$$a[\bar{1}01] \rightarrow \frac{a}{3}[\bar{2}11] + \frac{a}{3}[\bar{1}\bar{1}2] \quad (2.20)$$

- (v) dissociation by SISF and SESF on $\{111\}$ planes according to

$$a[\bar{2}11] \rightarrow \frac{a}{3}[\bar{1}2\bar{1}] + \frac{a}{3}[\bar{1}\bar{1}2] + \frac{a}{3}[\bar{2}11] + \frac{a}{3}[\bar{2}11] \quad (2.21)$$

- (vi) dissociation by APB on $\{010\}$ planes, according to

$$a[\bar{1}01] \rightarrow \frac{a}{2}[\bar{1}01] + \frac{a}{2}[\bar{1}01] \quad (2.22)$$

Extensive use has been made of transmission electron microscopy to study the dislocation dissociation reactions which occur in γ' . Scheme (v) has been observed in single-crystal alloys at temperatures of around 750 °C if the applied stress level is high enough to cause primary creep; this is considered in greater detail in Section 3.3 of Chapter 3. Of the other reactions given, no evidence has been found to support the dissociations given by (ii) and (iv) [36,37]. Dissociation by (i) and (iii) can occur when γ' or $\gamma + \gamma'$ alloys are deformed at the strain rates characteristic of a tension test. However, in practice, the dissociations are usually even more complicated than those given. This is because any given superdislocation is found to dissociate on both $\{111\}$ and $\{010\}$, with distinct segments of it lying on each of the two planes. For example, consider scheme (i). It is possible for the APB, or segments of it, to cross-slip to the cube plane $\{010\}$, whilst the two CSFs remain on $\{111\}$. This configuration is expected to be sessile, or 'locked', since the cube plane is not a glide plane for γ' at low temperatures. This configuration is termed a Kear–Wilsdorf lock, after those who discovered it in $L1_2$ crystal structures [38]. The behaviour of the Kear–Wilsdorf lock is central to our understanding of the anomalous yield effect displayed by $L1_2$ compounds and Ni_3Al in particular. This is considered in greater detail in Section 2.3.

C Point defects

The binary Ni–Al phase diagram indicates that the Ni_3Al compound is stable across a compositional range from approximately 23 to 27 at% Al. Thus, only a relatively small deviation from exact stoichiometry is possible, certainly very much smaller than that for the neighbouring β -NiAl compound, which displays the ordered B2 structure. Such deviations

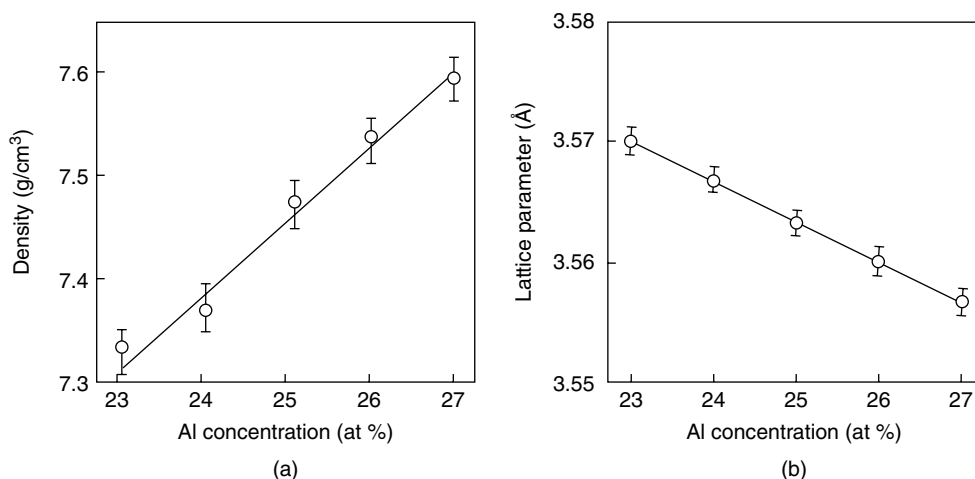


Fig. 2.41. Measurements [39] of (a) the density and (b) the lattice parameter with composition for the compound Ni_3Al .

from exact stoichiometry must be accommodated by point defects such as constitutional vacancies or anti-site defects. There are four distinct types of point defect which are conceivable, either (i) anti-site defects, i.e. Al atoms located on Ni sites, or alternatively Ni atoms on Al sites, or (ii) constitutional vacancies, i.e. vacancies on either the Ni or Al lattices. Studies have indicated that the Ni_3Al compound might more accurately be written as $\text{Ni}_{3+x}\text{Al}_{1-x}$ since non-stoichiometry is always accommodated by anti-site defects: Ni atoms on Al sites and Al atoms on Ni sites for Ni-rich and Ni-lean compositions, respectively. This is confirmed by measurements [39] of the density and lattice parameter, both of which vary monotonically as the composition passes through the point of exact stoichiometry; see Figure 2.41. In constructing a thermodynamic model for the Ni_3Al compound, Numakura *et al.* [40] have estimated that the density of constitutional vacancies at 950 K is very small, at $\sim 10^{-8}$ and $\sim 10^{-10}$ for the Ni and Al lattices, respectively, at the point of exact stoichiometry, varying by no more than an order of magnitude across the solubility range. Once again, this behaviour is in contrast to that of $\beta\text{-NiAl}$, for which Ni constitutional vacancies exist on the Al-rich side and Ni anti-site defects exist on the Ni-rich side [41].

The formation of anti-site defects in Ni_3Al has been inferred from the mechanical loss (internal friction) spectrum in $\text{Ni}_3(\text{Al}, \text{Ta})$ single crystals, by measuring the free decay of vibrations in a torsional pendulum [42]. The stresses set up in this way cause time-dependent, but recoverable, *anelastic* strains, due to the movement of the defects from one site to another. The internal friction shows a sharp maximum at frequencies between 0.03 Hz and 3 Hz, the exact value depending upon the temperature; see Figure 2.42. The activation energy is then about 3.0 eV, very much greater than that for the formation of a constitutional vacancy, but a reasonable value for the sum of the activation energies for vacancy formation and migration of an anti-site defect. Since the anelastic strains necessary for this effect must be due to a defect which breaks the cubic symmetry, the lattice strain introduced by a Ni atom on an Al site cannot be responsible since it is isotropic; on the other hand, an

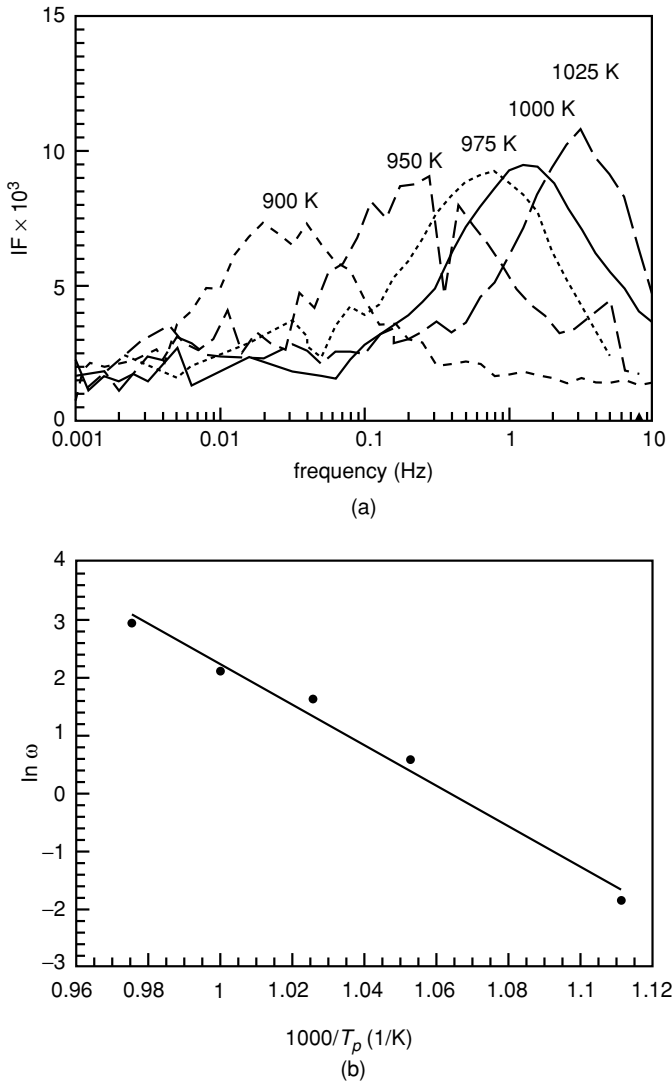


Fig. 2.42. Internal friction (IF) measurements [42] on polycrystalline $Ni_3(Al, Ta)$. (a) The variation of the IF signal as a function of frequency and temperature. (b) Arrhenius plot corresponding to (a), from which an activation energy H_{act} of 290 kJ/mol is deduced.

Al atom on a Ni site causes the tetragonal distortion that is required for the mechanical damping to occur.

2.3 Strengthening effects in nickel alloys

The mechanical properties of nickel alloys depend strongly upon the state of microstructure, which, in turn, is controlled by the chemical composition and the processing conditions.

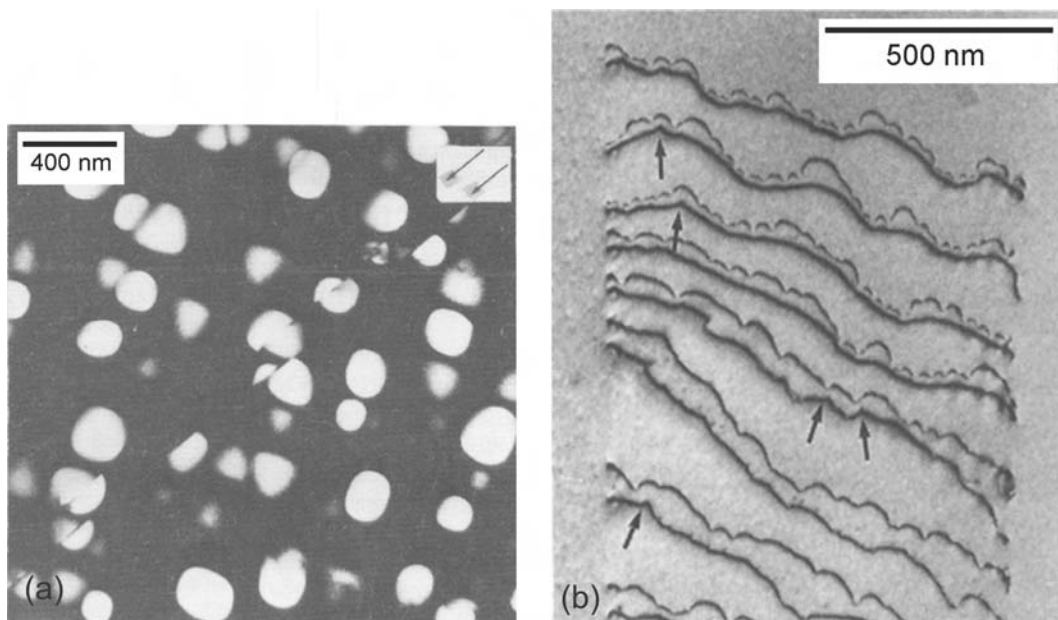


Fig. 2.43. Transmission electron micrographs [45,46] which support the view that dislocations travel through the γ/γ' microstructure in pairs: (a) dark-field micrograph of sheared particles in Nimonic 105, of size about 140 nm; (b) pair of edge dislocations in Nimonic PE16, $r = 8$ nm and $f = 0.09$, imaged after unloading.

This is particularly the case for properties such as the yield stress, which is a sensitive function of the distribution of the γ' phase – since this has a potent particle-strengthening effect. In this section, the factors which influence the strengthening by γ' are considered in detail.

2.3.1 Strengthening by particles of the gamma prime phase

It has been emphasised that a $a/2\langle 1\bar{1}0 \rangle\{111\}$ dislocation travelling in γ cannot enter the γ' phase without the formation of an anti-phase boundary (APB), and therefore that the dislocations must travel through the γ/γ' structure in pairs, with a second $a/2\langle 1\bar{1}0 \rangle\{111\}$ dislocation removing the anti-phase boundary introduced by the first. This is supported by evidence (see, for example, refs. [43]–[46]) from transmission electron microscopy (TEM); see Figure 2.43. The associated anti-phase boundary energy, γ_{APB} , represents a barrier which must be overcome if particle cutting is to occur. Although detailed calculations are required for an exact estimate, the particle-cutting stress is expected to be of the order γ_{APB}/b , where b is the Burgers vector. TEM experiments [47] indicate that $\gamma_{\text{APB}} \sim 0.1 \text{ J/m}^2$; consequently, with $b = 0.25 \text{ nm}$, the cutting stress is approximately $0.1/0.25 \times 10^{-9}/10^6$, or 400 MPa. This value is considerable, and it indicates that substantial *order strengthening* is expected. Indeed, in the nickel-based alloys, the contribution from order strengthening outweighs other contributions due to differences of modulus, stacking fault energy, interfacial energy and Orowan strengthening.

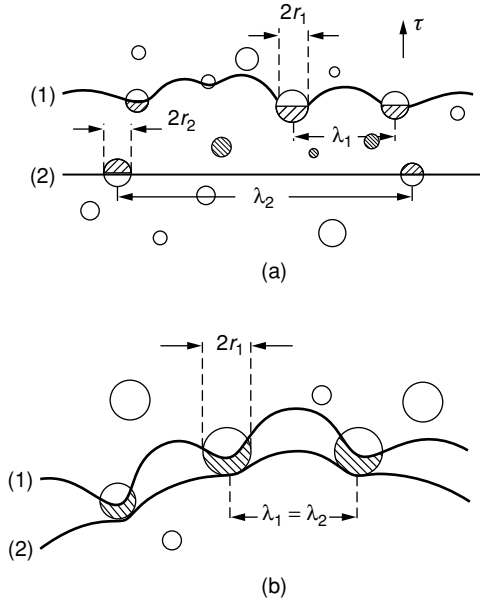


Fig. 2.44. Particles of ordered γ' being sheared by pairs of dislocations: (a) weak pair-coupling and (b) strong pair-coupling. Adapted from ref. [49].

A The case of weakly coupled dislocations

In order to develop expressions for the strengthening expected from γ' particles of volume fraction f , the case of 'weakly coupled' dislocations [48] is considered first; see Figure 2.44(a). By 'weak' one means that the spacing of the two paired dislocations is large in comparison with the particle diameter; consequently, the second trailing dislocation is some way behind the first, leaving faulted particles between the two. This situation is applicable to the case of the under-aged condition. To simplify matters, the γ' particles are assumed to be spherical and the volume fraction of γ' is taken to be small. The particles intersecting the first and second dislocations are assumed to be spaced at intervals λ_1 and λ_2 , respectively, and each is known as a 'Friedel spacing'. The radius of the particles along the first and second lines are r_1 and r_2 .

The onset of particle shearing is controlled by the following forces which act on the pair of dislocations: (i) the forces $\tau b\lambda_i$ ($i = 1, 2$) driving the particle-shearing process, due to the applied shear stress; (ii) the elastic force, R , repulsive in sign since the dislocations are of the same sense, acting to keep the pair separated; and (iii) the pinning forces, F_i ($i = 1, 2$), in magnitude equal to $2\gamma_{\text{APB}}r_i$, which are a consequence of the APB energy. The force balances may be written as follows [48,49]:

$$\tau b\lambda_1 + R + F_1 = 0 \quad (2.23a)$$

for the first dislocation and

$$\tau b\lambda_2 - R + F_2 = 0 \quad (2.23b)$$

for the second, where $F_1 = -2\gamma_{\text{APB}}r_1$ and $F_2 = +2\gamma_{\text{APB}}r_2$. The terms F_1 and F_2 are of opposite signs since the first dislocation introduces the APB, whereas the second dislocation removes it. The net shear stress required for cutting, denoted τ_c , is found by elimination of

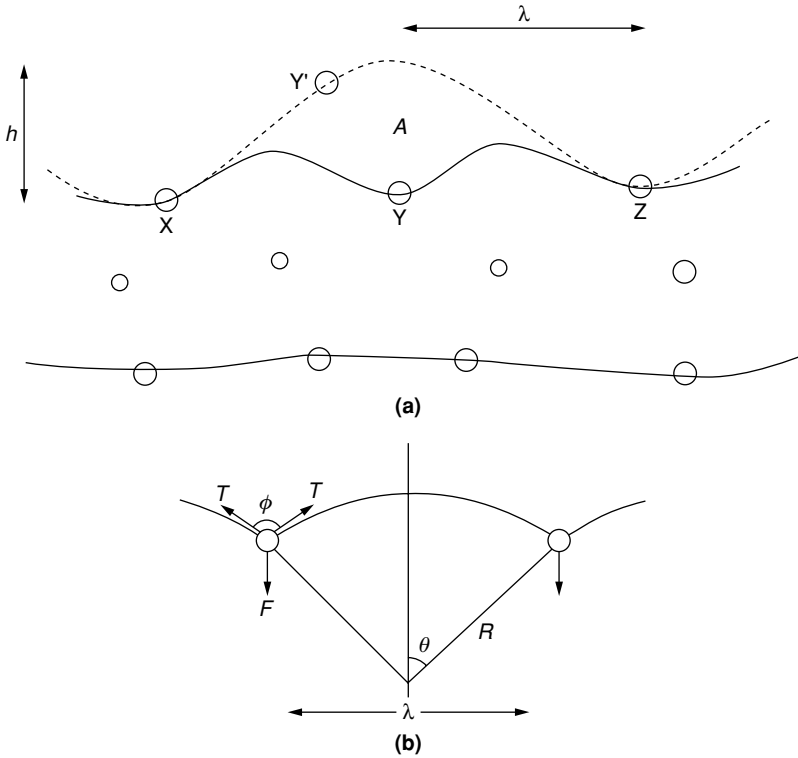


Fig. 2.45. For weakly coupled dislocations, (a) illustration of the unzipping of the leading dislocation from the pinning point Y to Y' – an area A on the slip plane is swept out; (b) geometry for force balance calculation between the dislocation line tension and pinning force due to order hardening, in which the particles are assumed to be point sources.

R from Equations (2.23a) and (2.23b); this yields [49]

$$\tau_c = \frac{\gamma_{APB}}{b} \left(\frac{2r_1 - 2r_2}{\lambda_1 + \lambda_2} \right) \simeq \frac{\gamma_{APB}}{2b} \left[\frac{2r_1}{\lambda_1} - \frac{2r_2}{\lambda_2} \right] \quad (2.24)$$

the second expression following from the approximation that λ_1 and λ_2 are expected to be very similar. Equation (2.24) indicates that τ_c depends critically upon the difference between the ratios $2r_1/\lambda_1$ and $2r_2/\lambda_2$, and therefore the way in which the dislocations interact with the γ' particles. The factor of $\frac{1}{2}$ in the first term of Equation (2.24) can be thought of as arising from the presence of the second dislocation.

Figure 2.43 indicates that the onset of particle shearing is controlled by the behaviour of the leading dislocation, which will, in practice, be held up by closely spaced, larger particles. It must bow in order to overcome the pinning forces due to the particles, i.e. to develop a component of the line tension in a direction opposite to that in which the pinning forces act. On the other hand, the trailing dislocation can be assumed to be straight; thus, to a good approximation, $2r_2/\lambda_2 = f$. The challenge therefore is to estimate the ratio $2r_1/\lambda_1$. This can be done by considering the conditions necessary for the leading dislocation to break away or 'unzip' from a particle; see Figure 2.45(a). The leading dislocation (initially

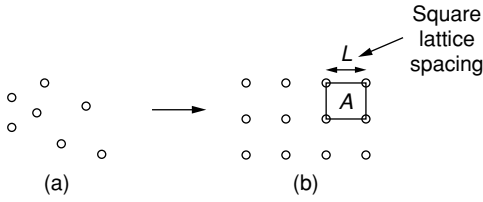


Fig. 2.46. Representation of a random distribution of γ' particles on the slip plane, (a) by a 'square lattice array', (b) such that the number density, N_s , is identical in both cases.

pinned at X, Y and Z) breaks away from Y and becomes pinned at Y', bowing out to another configuration compatible with the stress field and sweeping out an area A on the slip plane. The new configuration has a radius of curvature denoted R .

In practice, the curvature of the leading dislocation is caused by the resistance imparted by the particles – therefore the force balance in the vicinity of a particle needs to be considered carefully (see Figure 2.45(b)). Each particle is considered to be a point obstacle providing a pinning force $F = 2\gamma_{APB}r$, the largest that can be exerted, consistent with the 'unzipping' condition being analysed. The condition for static equilibrium is given by resolving forces in the vertical direction [48]; it is

$$2T \cos\{\phi/2\} = F \quad (2.25)$$

where $\phi = \pi - 2\theta$ and T is the line tension, which is approximately $\frac{1}{2}Gb^2$. The angles ϕ and θ are defined in Figure 2.45(b). Also by geometry

$$\lambda_1 = 2R \sin \theta \quad (2.26)$$

In order to proceed to an estimate of $2r_1/\lambda_1$, a relationship is required between λ_1 and the area, A , swept out by the unzipping process. To derive this, one models the random array of particles on the slip plane as a 'square lattice' of spacing L , such that $A = L^2$; see Figure 2.46. Then the number of particles per unit area, N_s , equals L^{-2} , where $L = A^{-1/2}$; the term N_s is equal to $N_V \times 2r$, where N_V is the number of particles per unit volume, consistent with $N_V = f/\frac{4}{3}\pi r_1^3$. It follows that

$$L = \left(\frac{2\pi}{3f}\right)^{1/2} r_1 \quad (2.27)$$

It is assumed that A can be approximated by $h\lambda_1$, where h is defined by the semicircle passing through X, Y' and Z (see Figure 2.45(a)); this has radius R . From the property of a circle, one has $R^2 = \lambda_1^2 + (R - h)^2 = \lambda_1^2 + R^2 - 2Rh + h^2$; if $h \ll R$, then $\lambda_1^2 = 2Rh$. Consequently,

$$\lambda_1^3 = 2Rh \times \frac{L^2}{h} = 2L^2R = L^2\lambda_1/\cos\{\phi/2\} \quad (2.28a)$$

and thus

$$\left(\frac{\lambda_1}{L}\right)^2 = \frac{1}{\cos\{\phi/2\}} = \frac{2T}{F} \quad (2.28b)$$

so that

$$\lambda_1 = \left(\frac{2T}{F} \right)^{1/2} \times L \quad (2.28c)$$

or, making use of Equation (2.27),

$$\lambda_1 = \left(\frac{2T}{F} \right)^{1/2} \times \left(\frac{2\pi}{3f} \right)^{1/2} \times r_1 \quad (2.28d)$$

Equations (2.28c) and (2.28d) are the critical equations for the Friedel spacing, λ_1 ; as expected, it is proportional to the square lattice spacing, L , which is, in turn, inversely proportional to the square root of the unzipped area, A .

An estimate of the critical resolved shear stress, τ_c , is derived by inserting Equation (2.28d) into Equation (2.24) and placing F equal to $2\gamma_{APB}r_1$. The result is as follows:

$$\tau_c = \frac{\gamma_{APB}}{2b} \left[\left(\frac{6\gamma_{APB}fr}{\pi T} \right)^{1/2} - f \right] \quad (2.29)$$

where r_1 has been replaced by r since the result will be valid for the general case of a uniform distribution of particles.

One can see that the factor $\frac{1}{2}[(6\gamma_{APB}fr/(\pi T))^{1/2} - f]$ modifies the first estimate of γ_{APB}/b for the cutting stress. It is instructive to introduce some numbers for the quantities in Equation (2.29). With $\gamma_{APB} \sim 0.1 \text{ J/m}^2$, $f = 0.3$, $r = 25 \text{ nm}$, $G = 80 \text{ GPa}$ and $b = 0.25 \text{ nm}$, the second factor $[(6\gamma_{APB}fr/(\pi T))^{1/2} - f]$ in Equation (2.29) is ~ 0.46 , i.e. not substantially different from unity. A final estimate of τ_c is therefore 90 MPa , a factor of ~ 4 different from the 400 MPa determined from the ratio γ_{APB}/b .

In practice, for large particles, $(6\gamma_{APB}fr/(\pi T))^{1/2} \gg f$ so that, to a good approximation, $\tau_c \propto (fr)^{1/2}$. Thus, significant fractions of larger particles are expected to promote hardening. Data [48] for the age hardening of monocrystalline Ni–Al alloys tested in compression have confirmed the $r^{1/2}$ dependence expected as $f \rightarrow 0$; see Figure 2.47. However, as $r^{1/2}$ becomes large, the linear dependence breaks down; there are a number of reasons for this behaviour. First, the point obstacle approximation is no longer valid, since the volume fraction, f , of γ' is then large. Second, the dislocation pairs become more strongly coupled; an analysis of the kind given in the following section is required.

B The case of strongly coupled dislocations

When the γ' particles are large, as will be the case for the over-aged condition, the spacing of the dislocation pairs becomes comparable to the particle diameter. Thus, any given particle may contain a pair of dislocations which are now ‘strongly coupled’; see Figure 2.44(b). In this case, the behaviour becomes critically dependent upon the elastic repulsive force, R , since this must be overcome if the trailing dislocation is to enter the γ' particle. The analysis proceeds as follows [49]. The Friedel spacings, λ_1 and λ_2 , are now equal, and can be approximated by the square lattice spacing, L . In these circumstances, Equations (2.23a) and (2.23b) can be manipulated to yield [49]

$$2\tau bL + (F_1 + F_2) = 0 \quad (2.30a)$$

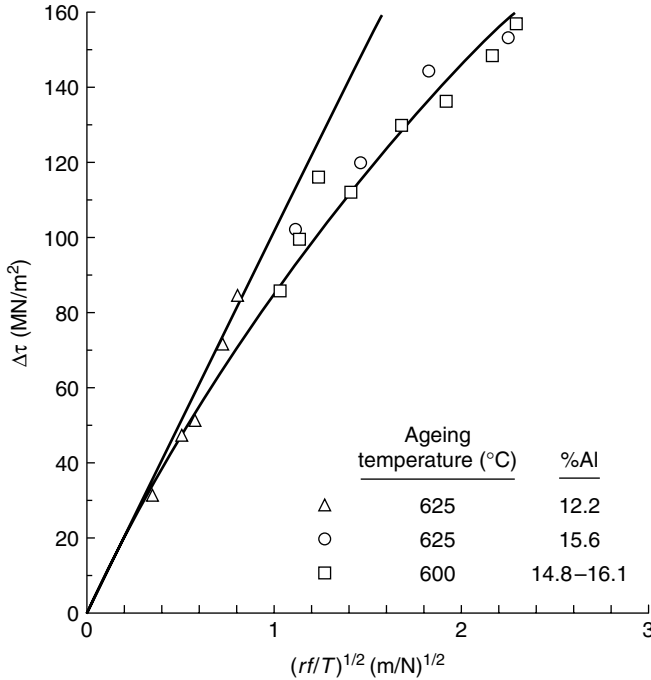


Fig. 2.47. Illustration of the agreement between the degree of ordering hardening, $\Delta\tau$, measured during room-temperature compression testing of Ni–Al single crystals and $(rf/T)^{1/2}$, where r is mean particle radius, f is volume fraction and T is dislocation line tension [48].

and

$$2R + (F_1 - F_2) = 0 \quad (2.30b)$$

Thus, $\tau = -\frac{1}{2}(F_1 + F_2)/(bL)$, where both F_1 and F_2 depend critically upon the line length, $\ell\{x\}$, of the corresponding dislocation inside the particle; see Figure 2.48. The symbol x denotes the distance to which a particle is penetrated, measured from the point of entry. Elementary geometry indicates that the function $\ell\{x\}$ is given by $2(2rx - x^2)^{1/2}$, if the particles are spherical. The forces F_1 and F_2 are of opposite sign; therefore force–distance plots (see Figure 2.49) confirm that the maximum value of $(F_1 + F_2)$ occurs when the trailing dislocation just touches the particle, at which point $x = x_m$ defines the position of the leading dislocation. Then $F_2 = 0$, and Equations (2.30a) and (2.30b) reduce to

$$2\tau bL = -F_1\{x_m\} = \gamma_{APB}\ell\{x_m\} = 2R \quad (2.31)$$

Thus one can see that the magnitude of the repulsive force R determines the critical value of τ required for particle shearing.

An estimate of R can be made by appealing to dislocation theory [50]. The repulsive force acting between two dislocations of the same sign, separated by a distance x_m , is approximately $Gb^2/(2\pi x_m)$ per unit length of the dislocation line. To account for small uncertainties, a dimensionless constant, w , is introduced which is expected to be of the

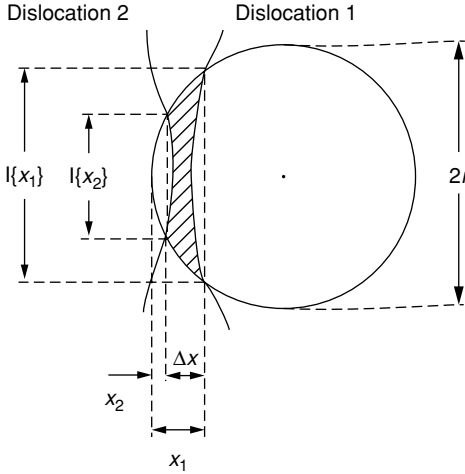


Fig. 2.48. Representation of the strong interaction between a pair of dislocations and a γ' particle, assumed to be spherical. The hatched area denotes an APB. Adapted from ref. [49].

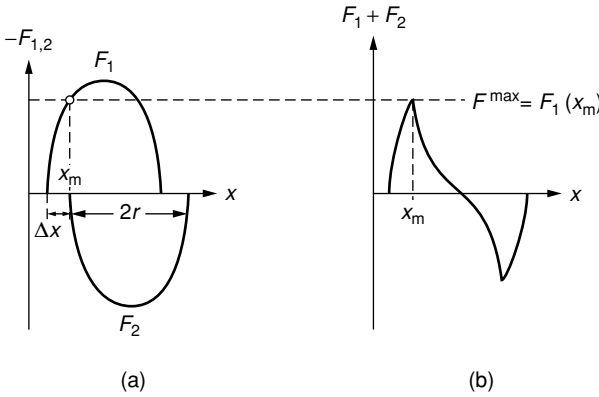


Fig. 2.49. Force–distance profiles of dislocations cutting an ordered γ' particle, assumed spherical: (a) variation of the forces F_1 and F_2 acting on the first and second dislocations with distance x , measured from the point of penetration; (b) net force on the pair, their spacing assumed to be constant. Adapted from ref. [49].

order of unity; R is then approximately given by

$$R = w \left(\frac{Gb^2}{2\pi} \right) \frac{\ell\{x_m\}}{x_m} \quad (2.32)$$

A first estimate of τ for the case of strongly coupled dislocations is derived by combining Equations (2.31) and (2.32), yielding

$$\tau = \frac{1}{2} \left(\frac{Gb}{L} \right) \frac{w}{\pi} \frac{\ell\{x_m\}}{x_m} \quad (2.33)$$

Clearly, an estimate of the spacing x_m is required. It follows also from Equations (2.31) and (2.32), giving

$$x_m = \frac{2w}{\gamma_{\text{APB}}} \left(\frac{Gb^2}{2\pi} \right) \quad (2.34)$$

so that [51]

$$\tau_c = \frac{1}{2} \left(\frac{Gb}{L} \right) \frac{2w}{\pi} \left(\frac{2\pi\gamma_{APB}r}{wGb^2} - 1 \right)^{1/2} \quad (2.35)$$

Finally, one can substitute into Equation (2.35) the expression for L given by Equation (2.27), to identify the dependence upon r and f ; this is

$$\tau_c = \sqrt{\frac{3}{2}} \left(\frac{Gb}{r} \right) f^{1/2} \frac{w}{\pi^{3/2}} \left(\frac{2\pi\gamma_{APB}r}{wGb^2} - 1 \right)^{1/2} \quad (2.36)$$

A number of interesting points arise from this analysis. First, the factor of $\frac{1}{2}$ arising in Equation (2.35) can again be traced to the pair-coupling of the dislocations. Taking $\gamma_{APB} \sim 0.1 \text{ J/m}^2$, $f = 0.3$, $r = 25 \text{ nm}$, $G = 80 \text{ GPa}$ and $b = 0.25 \text{ nm}$, as before, one finds that $\tau_c = 141 \text{ MPa}$. This is about 50% lower than the Orowan bowing stress given by Gb/L , which features in the expression for τ_c in Equation (2.35) – suggesting that the Orowan mechanism is unlikely to be the major cause of strengthening in nickel alloys, at least at room temperature. Finally, as $r \rightarrow \infty$, an $r^{-1/2}$ dependence for τ_c is expected, so that the resistance provided by strongly coupled dislocations disappears at large particle sizes. This is in contrast to the $r^{1/2}$ dependence for weakly coupled dislocations, which is found provided that f is small; clearly in that case the strengthening increases monotonically as r increases. In fact, it can be shown that the transition from ‘weak’ to ‘strong’ coupling occurs when $r \sim 2T/\gamma_{APB}$, yielding a peak-strength of $\sim \frac{1}{2}\gamma_{APB}f^{1/2}/b$.

That the optimum hardening in the nickel-based superalloys occurs for a γ' particle size which lies at the transition from weak to strong coupling has been confirmed by a number of studies. Reppich *et al.* [43,44,52] have measured the critical resolved shear stress at room temperature for the nickel-based superalloys PE16 and Nimonic 105, aged such that the volume fraction, f , and mean particle diameter, r , varied substantially. These systems show a degree of solid-solution strengthening, estimates for which were subtracted from the total strengthening displayed. The results are given in Figure 2.50. For PE16, an optimum r was found to be in the range 26–30 nm. For Nimonic 105, the range was found to be 55–85 nm. Jackson and Reed [51] have used these concepts to design a heat treatment schedule for the Udimet 720Li superalloy which places the γ' particles at a size which confers hardening at the transition from weak to strong coupling – and thus the maximum which can be achieved.

2.3.2 Temperature dependence of strengthening in the superalloys

The nickel alloys exhibit a remarkable characteristic: the yield stress, by which one means the stress necessary to cause the onset of plastic deformation by dislocation flow, does not decrease strongly with increasing temperature, as is the case for most other alloy systems. In fact, for many superalloys the yield stress *increases* with increasing temperature, typically until temperatures of about 800 °C are reached. Figure 2.51 shows some typical data for a number of single-crystal alloys, tested along the $\langle 001 \rangle$ direction. The peak stress of beyond 1000 MPa is about 50 times greater than the flow stress of pure Ni – which confirms that considerable metallurgical strengthening effects are at play. For temperatures beyond 800 °C,

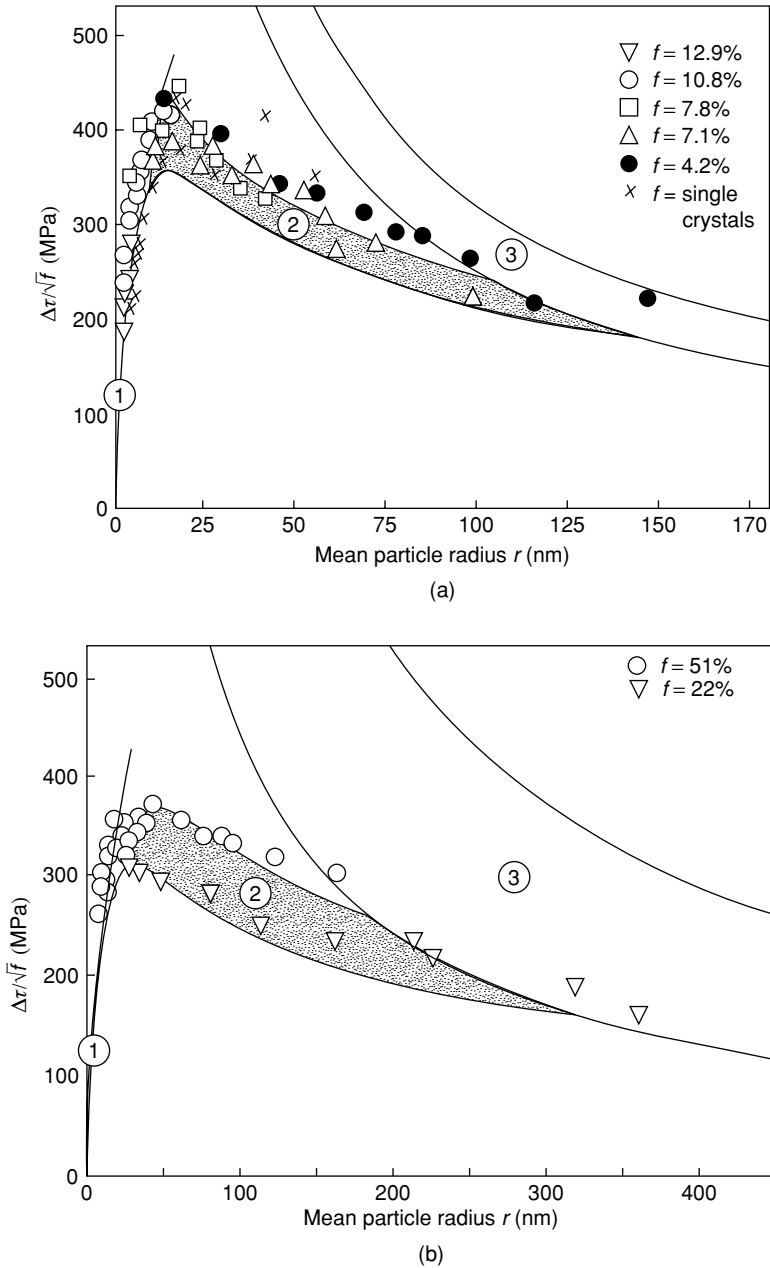


Fig. 2.50. Data [52] for the increase in critical resolved shear stress (divided by the square root of the volume fraction of the particles) as a function of the mean particle radius for two nickel-based superalloys: (a) Nimonic PE16 ($\gamma_{APB} = 0.125 \text{ J/m}^2$) and (b) Nimonic 105 ($\gamma_{APB} = 0.110 \text{ J/m}^2$). In regime 1, hardening is by weakly coupled pairs; in regime 2 it is by strongly coupled pairs; and in regime 3 it is by the Orowan mechanism. Note that the peak strength is associated with the transition from weak to strong coupling.

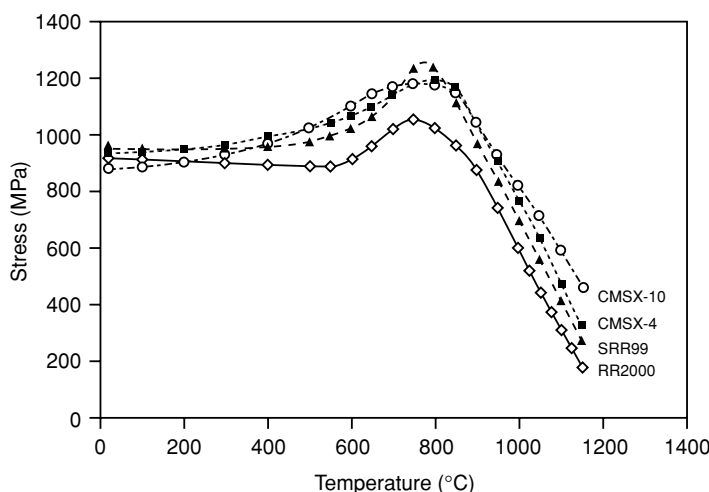


Fig. 2.51. Variation of the yield stress of a number of single-crystal superalloys with temperature.

the yield stress decreases quickly; by about 1200 °C little resistance to plastic deformation is displayed.

In a first attempt to rationalise this behaviour, Davies and Stoloff [53] studied the yield properties of Ni–Al binary alloys between –200 °C and 800 °C; see Figure 2.52. The Ni–8%Al alloy, which is alloyed insufficiently for the precipitation of the γ' phase, thus benefits only from a solid-solution strengthening of between 50 and 100 MPa, depending upon the temperature. The Ni–14%Al alloys displayed precipitation strengthening, but to an extent dependent upon the fraction, size and distribution of the γ' particles, and hence on the heat treatment applied. Quenching from 1000 °C introduced only very fine γ' precipitates which were barely resolvable in the transmission electron microscope (TEM). Ageing at 850 °C introduced particles spaced by 200–250 nm, significantly coarser than the 40–60 nm range when heat treatment was carried out at 700 °C. The TEM microscopy indicated that the micromechanism of deformation was Orowan looping when ageing was carried out at 850 °C, but particle cutting was observed for material subjected to the 700 °C treatment. This is strong evidence that the rather constant yield stress displayed by the Ni–14%Al alloy heat treated at 700 °C is due to microscopic deformation of γ' .

Further work reported by Pearcey *et al.* [54] sheds light on the role of γ' in the deformation of the superalloys. The Mar-M200 alloy, which contains about 60% of γ' , was tested in single-crystal form (see Figure 2.53); this revealed a curve very similar to those of the more modern single crystals given in Figure 2.51. Also tested was single-crystal cube-orientated Ni₃Al, alloyed such that its composition matched that found in Mar-M200. The results (see Figure 2.53) confirm that at and beyond the peak stress, the behaviour of the alloy is determined by the strength of the γ' phase. Furthermore, from ambient to the temperature associated with the peak stress, the γ' imparts an increasing fraction of the strength as the temperature increases. Similar experiments were carried out by Beardmore *et al.* [55]. The temperature dependence of the yield stress was determined for a number of alloys with varying amounts of γ' . Rather ingeniously, a series of alloys lying on the tie-line

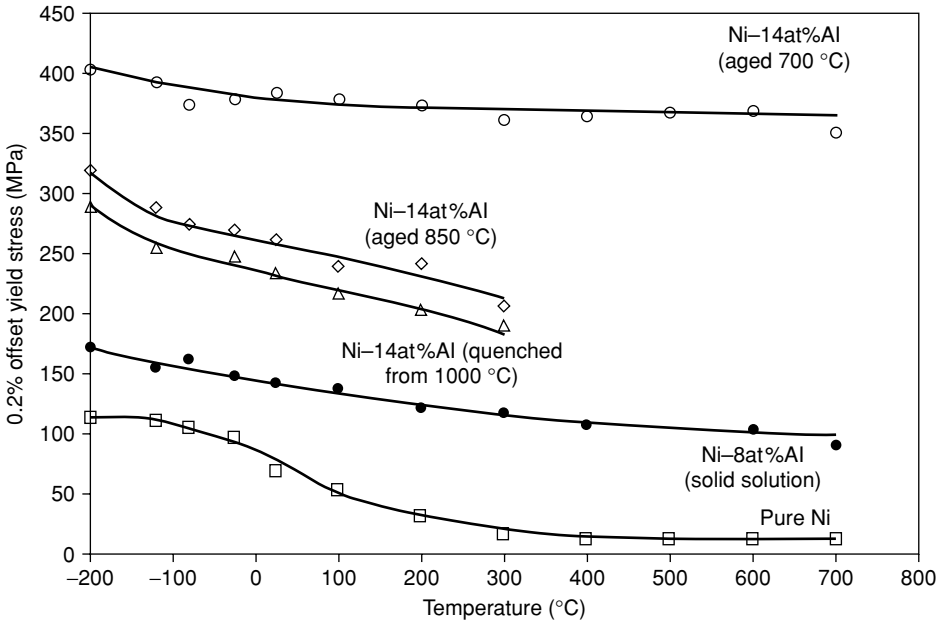


Fig. 2.52. Data for the yield stress of alloys in the binary Ni–Al system as a function of composition and temperature. Adapted from ref. [53].

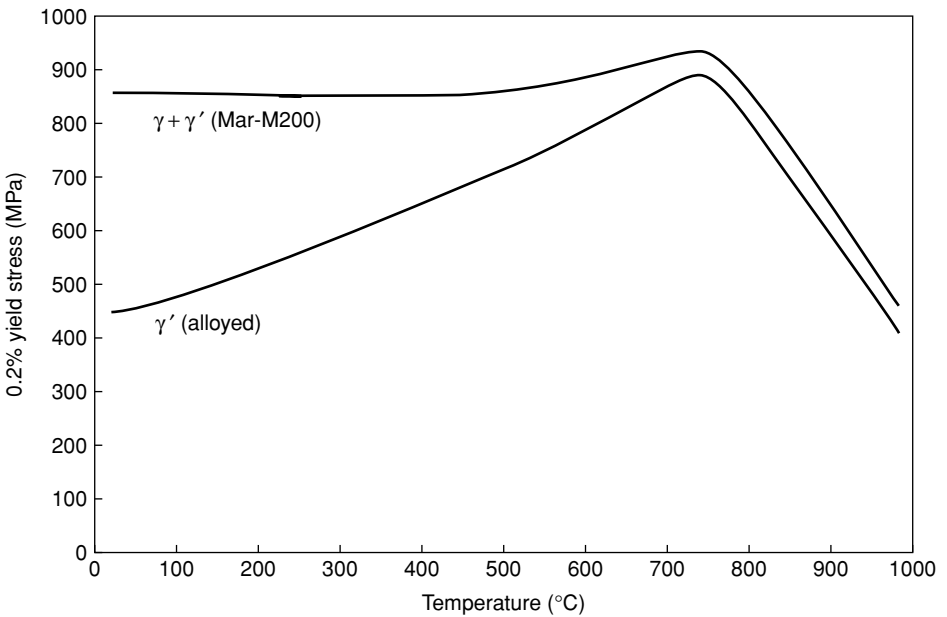


Fig. 2.53. Variation with temperature of the yield stress of the Mar-M200 alloy in single-crystal form, and a monolithic Ni_3Al alloy of composition equivalent to the γ' particles in it. In both cases, testing was along $\langle 100 \rangle$. Adapted from ref. [54].

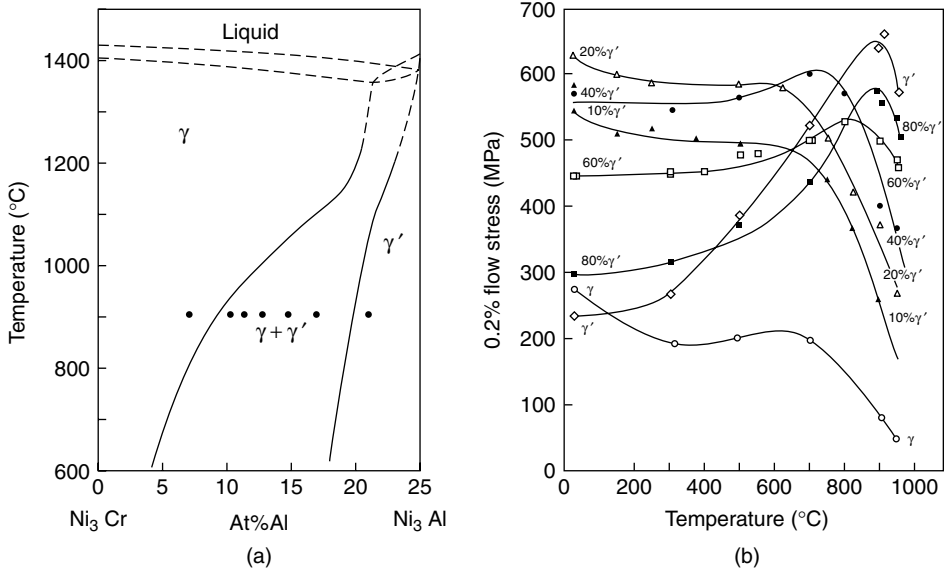


Fig. 2.54. (a) Section through the Ni–Cr–Al ternary phase diagram at 75 at% Ni, showing the alloys used in experiments by Beardmore *et al.* [55], and (b) variation of the yield stress with temperature for the alloys so produced.

joining Ni₃Al and the Ni–25at%Cr alloy were considered; see Figure 2.54(a). Once again, the alloy consisting of 100% γ' displayed a strong positive dependence on the temperature, until about 800 °C; see Figure 2.54(b). Above the temperature corresponding to the peak stress, the yield stress of a two-phase $\gamma + \gamma'$ alloy obeys a rule of mixtures, i.e. it corresponds to the weighted average of the values for γ and γ' . This contrasts strongly with the behaviour at low temperatures, where two-phase alloys are very much stronger than the rule of mixtures would predict.

The anomalous yielding effect is discussed in detail in the following section, but, broadly speaking, it arises for the following reason. Upon deformation, the applied stress, the anisotropy of the anti-phase boundary energy and further contribution from the elastic anisotropy combine to promote the *cross-slip* of segments of the γ' superpartial dislocations from the {111} slip plane to the cross-slip plane {001}. The force per unit length of dislocation which promotes cross-slip is equal to [56]

$$\gamma_{111} \left[\frac{1 + f_1 \sqrt{2}}{\sqrt{3}} \right] - \gamma_{100} \quad (2.37)$$

where γ_{111} and γ_{100} are the APB energies on the octahedral and cube planes, respectively, and f_1 is given by

$$f_1 = \sqrt{2} \left[\frac{A - 1}{A + 2} \right] \quad (2.38)$$

where A is Zener's anisotropy factor, defined according to

$$A = \frac{2C_{44}}{C_{11} - C_{12}} \quad (2.39)$$

where the C_{ij} 's are entries in the stiffness matrix; see the Appendix to this chapter. The cross-slipped segments represent *microstructural locks* since they resist deformation – they cannot move without trailing APBs behind them and are thus immobile. These are known as Kear–Wilks locks after the researchers who first proposed their existence [57]. The hardening is increasingly prevalent as the temperature rises, due a component of the cross-slipping process which is thermally activated. Beyond the peak stress, which occurs typically around 800 °C, slip-line trace analysis on deformed single crystals has revealed that the mode of slip deformation changes to $a/2\langle 1\bar{1}0\rangle\{001\}$, i.e. the so-called cube slip dominates [58].

2.3.3 The anomalous yielding effect in gamma prime alloys

It is now widely accepted that the cross-slip of dislocations from the octahedral to the cube plane – causing the formation of Kear–Wilks locks – is the root cause of the positive temperature dependence of the yield stress, and that a preference for the thermally activated slip on the cube plane is responsible for the softening which occurs beyond the temperature corresponding to the peak stress. Unfortunately, this picture of the deformation behaviour is incomplete in many ways, since it does not explain some important observations that have emerged from recent research carried out on this topic [59,60]. In particular, for most $L1_2$ -based nickel alloys the yield stress increases with temperature from about 77 K [61] – too low for a phenomenon which relies purely upon thermal activation. Furthermore, the behaviour is found to be reversible; thus, prior deformation at an elevated temperature exerts only a very modest effect on the low temperature behaviour, and only a small component of the work hardening increment at the higher temperature is inherited. These observations are consistent with a strain-activated phenomenon; thus cross-slip is obviously heavily influenced by the process of deformation. The work hardening rate is also found to be abnormally high – at a strain of 0.2%, it is one or two orders of magnitude greater than the value typically observed for single slip in FCC metals – and it increases in magnitude over some, if not all, of the temperature range of the anomaly. Furthermore, the strain-rate sensitivity is not at all pronounced, as would be expected for a diffusion-controlled phenomenon [62]. Other interesting effects include a significant tension/compression asymmetry, the magnitude of which varies with crystallographic orientation, so that Schmid's law is violated [63,64]. For single crystals aligned along $\langle 001 \rangle$, the critical resolved shear stress (CRSS) on $[\bar{1}01]\{111\}$ is greater in tension than in compression; conversely, for alignment at orientations lying along the $[011]/[\bar{1}11]$ boundary of the stereographic triangle, the CRSS is greater in compression rather than tension; see Figure 2.55. Such effects have been rationalised [65] by considering combinations of Schmid factors on the cube cross-slip (for the direction $\langle 110 \rangle$) and on the primary octahedral plane (along $\langle 110 \rangle$ and $\langle 112 \rangle$).

Microstructural observations using transmission electron microscopy have helped to clarify the micromechanical effects occurring during deformation, although one should note that considerable controversy still exists over their interpretation [60]. Post-mortem examinations reveal that long, straight screw superdislocations are prevalent in samples deformed in the anomalous regime, indicating that screws are very much less mobile than non-screws. The dislocation microstructure is remarkably homogeneous – there is little apparent tendency to form the sub-grains or cell structures found in pure metals [66] – so

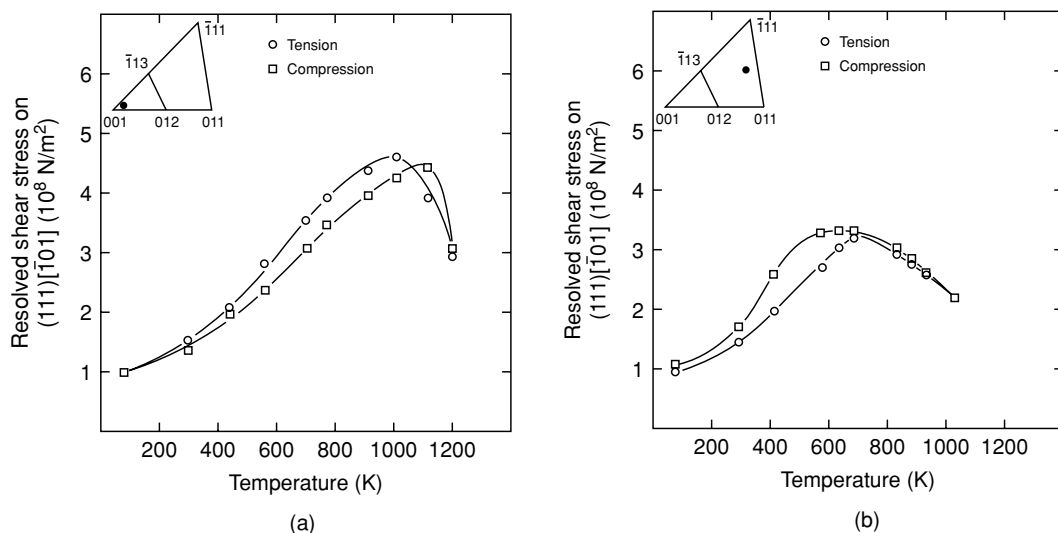


Fig. 2.55. Variation with temperature of the critical resolved shear stress for $[\bar{1}01]\{111\}$ slip for $\text{Ni}_3(\text{Al}, \text{Nb})$ single crystals, measured in tension and compression, for two orientations within the stereographic triangle [63]; note the considerable tension/compression asymmetry which violates Schmid's law.

that it seems likely that the macroscopic properties are controlled to a good approximation by the behaviour of individual dislocations. The superdislocations are indeed found to be dissociated completely onto the cube plane, i.e. into the Kear–Wilsdorf (KW) configuration, but incomplete KW locks are also present in which a portion of the APB resides on the octahedral plane. Incomplete KW locks, which are mechanically unstable and thus prone to relax back to the complete configuration, are preferred, particularly after deformation at low temperatures. *In-situ* TEM observations using straining stages have unambiguously confirmed the dynamical locking of screw dislocations by cross-slip. However, instead of being curled between pinning points, as might have been expected, mobile superpartials are found to lock themselves over significant lengths almost instantaneously and to remain rather rectilinear during the locking and unlocking processes; when dissociated on the octahedral plane their velocity is extremely high, but their motion is jerky. Veyssiere and Saada [60] have constructed diagrams which are helpful in visualising the situation; see Figure 2.56. As the leading superpartial cross-slips onto the cube plane, it does so by the lateral spreading of a pair of jogs in opposite directions; this must occur with the lateral displacement of a pair of macrokinks on the octahedral plane. It has been pointed out that the macrokinks can be of two types – simple macrokinks and switch-over macrokinks – the distinction depending upon whether a single superpartial remains consistently the first in a ribbon built up by multiple cross-slip events; see Figure 2.57. The situation then resembles the images [66] seen in the TEM; see Figure 2.58. Careful analyses of the distributions of the heights of macrokinks have been made as a function of the temperature of deformation [67]; heights no greater than ten times the APB width, λ_0 , are observed, the frequency increasing exponentially as the height reduces to zero, apart from a strong preference for heights of

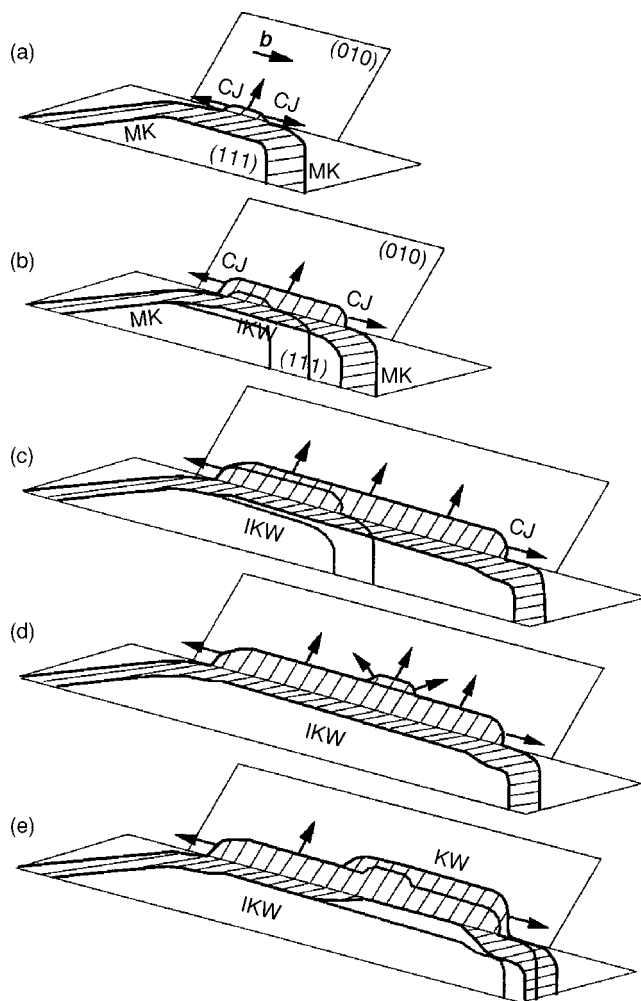


Fig. 2.56. Steps in the transformation of a screw superdislocation by cross-slip onto the cube plane [60]. CJ stands for complex jog and MK stands for macrokink; these lie on the cube and octahedral planes, respectively. KW and IKW are complete and incomplete Kear–Wilford segments, respectively, a distinction which depends upon whether the superdislocation is totally or partially dissociated on the cube plane.

one to two λ_0 . At this height, they are christened ‘elementary kinks’. Other research has emphasised the dissociation of the leading and trailing superpartials and thus the formation of complex stacking faults (CSFs); estimates of the CSF fault energy have been possible through the use of image simulation techniques. A strong influence of Hf on the CSF energy has been reported [47], which explains the effect of Hf alloying on the values of the peak temperature, stress and work hardening rate. The elements Ti and Ta behave similarly.

An adequate quantitative model for the macroscopic yielding behaviour of Ni_3Al which accounts for the macroscopic and microscopic observations – especially the influence of alloying – is not available as yet. In constructing it, one will need to decide whether the

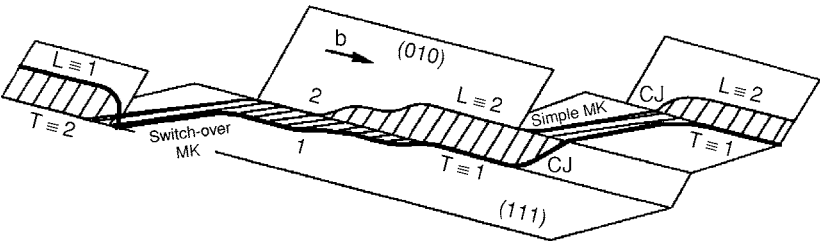


Fig. 2.57. Schematic representation of a ribbon of cross-slipped superdislocations, displaying macrokinks of both the simple and switch-over types [60] – the distinction depends upon whether the leading superpartial switches to the trailing one on either side of the cross-slipped segment.

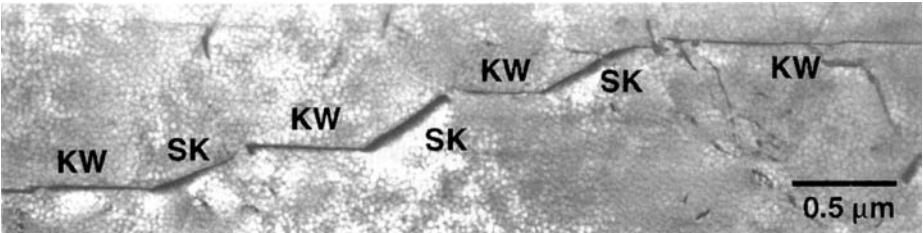


Fig. 2.58. Transmission electron micrograph [66] of a typical superdislocation in the deformed Ni₃Al compound. KW and MK refer to Kear–Wilsdorf locks and macrokinks, respectively.

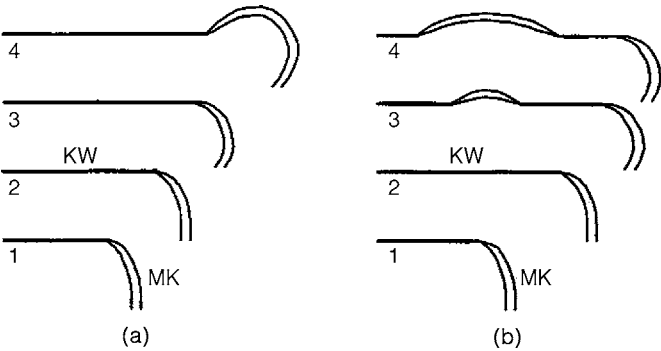


Fig. 2.59. Schematic illustration of the methods by which a KW lock may be overcome [60]: (a) bypassing by an expanding macrokink (MK), and (b) unlocking by the formation of a glissile double-kink configuration on the locked KW segment.

improvement in the flow stress with increasing temperature is due to an increase in the strength of the obstacles, or else to a reduction in their spacing; moreover, the extent to which the interaction of superdislocations is responsible for the work hardening behaviour (if at all) needs to be established. One can say that a successful theory is likely to be built on one or two foundations (see Figure 2.59): either (i) a treatment of locking/unlocking by an ‘unzipping’ process on the scale of the cores of the superpartial dislocations, which seems most likely to explain the considerable composition dependence displayed by the Ni₃Al compounds doped with Hf, Ta, Ti, etc. and ones away from the stoichiometric composition, and

(ii) treatments which emphasise the bypassing of obstacles by the propagation of macrokinks. The motivated reader is referred to ref. [60] for a full critique of the models developed so far. The approach proposed by Hirsch [68], which focusses attention on the macrokinks, is one of the more physically realistic models developed so far. Here, the height of the macrokinks, h , is related to the probability of locking per unit time, P_l , by the relationship $h = v_f/P_l$, where v_f is the free-flight velocity of glissile dislocations – this is expected to be proportional to the applied stress, σ . The probability of locking is then given by $P_l \propto l \exp\{-G_l/(kT)\}$, where G_l is an activation energy of locking and l is the spacing of macrokinks along the dislocation line; since this is assumed to vary as the inverse of σ , it follows that $h \propto \sigma^2 \exp\{+G_l/(kT)\}$. The *average* dislocation velocity, v , which is governed by the movement of macrokinks and thus the difference between the rates of pinning and unpinning events, is then given by

$$v = v_0 \left[\exp \left\{ -\frac{G_{ul}}{kT} \right\} - \exp \left\{ -\frac{G_l}{kT} \right\} \right] = v_0 \exp \left\{ -\frac{G_{ul} - G_l}{kT} \right\} \quad (2.40)$$

where v_0 is a constant, and G_{ul} is the activation energy for unlocking; this is assumed to depend upon the effective stress, σ , necessary to cut pinning points, taken to be the ends of the macrokinks; thus

$$G_{ul} = {}^0G_{ul} - \sigma V^* \quad (2.41)$$

where ${}^0G_{ul}$ is the activation energy in the absence of the applied stress and V^* is an activation volume – this is of order hb^2 , since the breaking of pinning points sweeps an area which is proportional to h . Note that it is expected that $G_{ul} > G_l$ so that the rate of locking will be greater than the rate of unlocking. If one ignores the small contribution of the Orowan stress, G_b/h , one has, by combining the various terms,

$$\sigma \propto [{}^0G_{ul} - kT \ln\{v_0/v\} - G_l]^{1/3} \exp \left\{ -\frac{G_l}{3kT} \right\} \quad (2.42)$$

so that the increase in σ with T is reproduced by the exponential term. The factor of 3 present in it is notable; it arises because of the assumed inverse square dependence of h on σ . Unfortunately, whilst a decrease in h with increasing σ is a reasonable proposition, this dependence has yet to be confirmed experimentally.

2.4 The creep behaviour of nickel alloys

So far, it has been assumed that a unique value of the plastic strain is produced when a nickel alloy is loaded under any given conditions of temperature and applied stress. In practice, this is not precisely the case – particularly at high temperatures, plastic strain is accumulated over time by the process of creep. This effect is pertinent to the turbine blading in the aeroengine, which is machined to tight tolerances with respect to the engine housing; consequently, excessive creep deformation during service constitutes ‘failure’ of the component since it is no longer able to satisfy the function for which it was designed. It follows that creep-resistant alloys are required for this application.

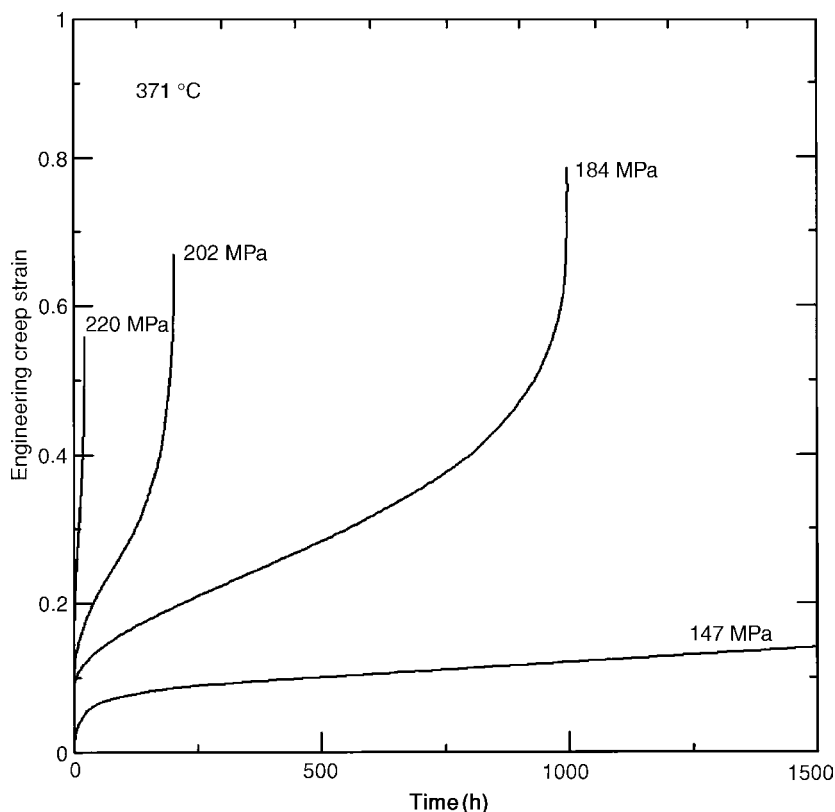


Fig. 2.60. Creep curves (constant load conditions) for nominally pure nickel tested at 371 °C [69].

2.4.1 The creep behaviour of nickel

Creep strengthening in polycrystalline nickel alloys arises both from solid-solution strengthening due to the presence of solute atoms and from precipitation hardening due to phases such as γ' . Consider Figure 2.60, in which creep data [69] are given for nominally pure nickel tested under constant load conditions at 371 °C and stresses between 147 and 220 MPa. It is clear that the evolution of creep strain is remarkably sensitive to the stress applied. Moreover, for each creep curve a rapid initial transient (known as a period of *primary* creep) gives way to a regime in which, to a good approximation, the creep strain rate is constant – this steady-state is referred to as *secondary* creep. The onset of failure is associated with a divergence from the steady-state behaviour; during this *tertiary* regime the creep strain rate increases with increasing strain before fracture finally occurs. Figure 2.61 shows additional creep data [70] for polycrystalline nickel dispersion-strengthened with a small fraction of ThO_2 particles, under constant stress conditions of 28 MPa over a range of temperatures. One can see that the creep strain is sensitive to the temperature employed – a rise of 25 °C is sufficient to double the creep strain. The fitting of the steady-state creep data in ref. [70] to Equation (1.7) yields an estimate of the activation energy for creep, $Q = 270 \text{ kJ/mol}$, which is close to the value for the self-diffusion of nickel. When one collates the various

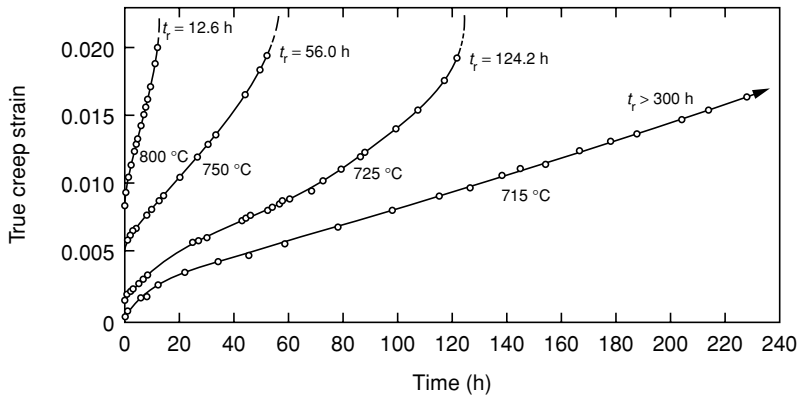


Fig. 2.61. Creep curves for nickel doped with a small fraction of ThO_2 particles [70]; $\sigma = 28 \text{ MPa}$. The time to rupture, t_r , is given for each creep curve.

creep data which are available for pure nickel [71], one finds that a stress exponent, n , of about 5 fits the data reasonably well. This fifth-power law dependence on the applied stress is commonly found in pure metals [72,73]; see Figure 2.62.

The steady-state creep behaviour of nickel can be rationalised by assuming that the modes of softening and hardening, i.e. dislocation annihilation and multiplication, are operating at rates which are balanced so that the dislocation density is constant. The various ‘recovery-controlled’ creep models available in the literature are based upon this premise [73–75]. Then, provided that the temperature is sufficiently high, creep occurs in pure metals such as nickel by a combined climb-plus-glide mechanism – if a gliding dislocation becomes pinned, then a small amount of climb can release it, allowing further glide to occur. Dislocation glide is then responsible for the majority of the deformation, although the average velocity is determined by the climb step [74]; for most metals creeping above $0.6T_m$, climb occurs by lattice-diffusion (rather than, for example, by pipe diffusion along a dislocation core), and therefore this process is rate-controlling. In the steady-state regime, well-established dislocation cell sub-structures with associated sub-grain boundaries are observed using transmission electron microscopy – within the grain boundaries; see Figure 2.63. In general, the formation of sub-grains arises principally as a result of polygonisation due to the climb of edge dislocations since cross-slip is easy [72] – studies suggest that the sub-grain size and misorientation reach a constant value during primary creep consistent with the strain rate arising. The sub-grain size is found to be inversely proportional to the applied stress; moreover, the same sub-grain size is developed in steady-state creep, regardless of its previously work hardened or pre-crept condition. If the steady-state regime is interpreted in this way, then primary creep arises because a constant sub-structure is yet to be established; the rate of dislocation multiplication then outweighs the rate of the annihilation processes. Conversely, during tertiary creep other degradation mechanisms are occurring – for example, loss of net cross-section due to creep cavitation at grain boundaries.

Nevertheless, the σ^5 -dependence of the steady-state strain rate needs to be rationalised. This can be done in the following way, the analysis being inspired by the work reported in refs. [76] and [77]. Assume that the characteristic dimension of the dislocation sub-structure

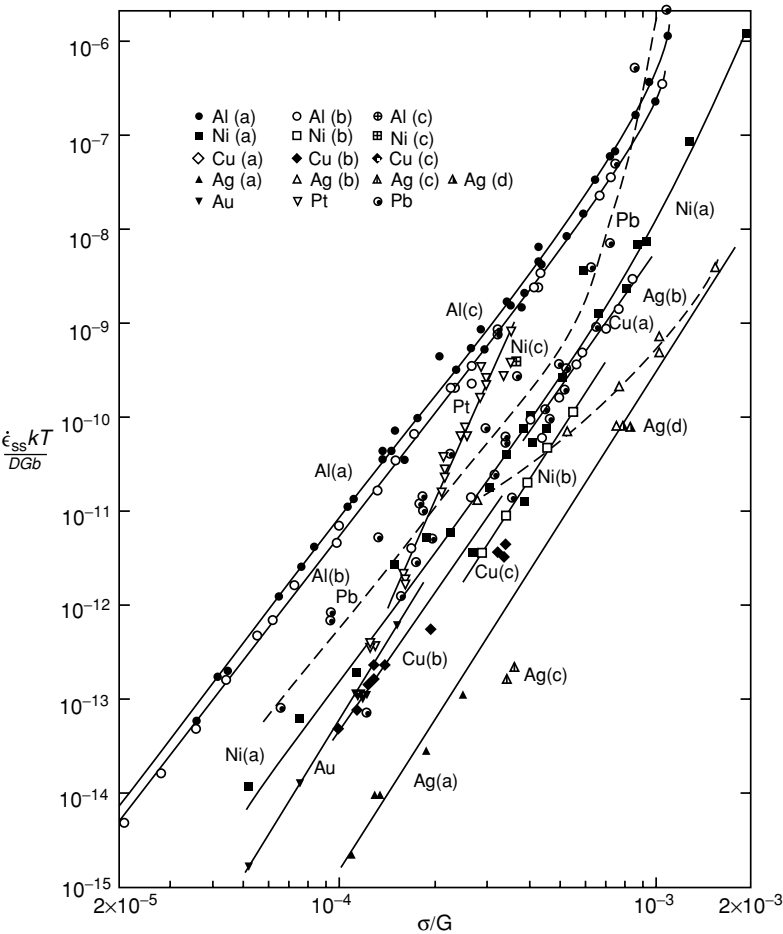


Fig. 2.62. Effect of stress on the normalised steady-state creep rates of the face-centred cubic (FCC) metals [72]. Note that the Al(b), Ni(a), Ag(a) and Ag(d) tests were not performed at constant stress.

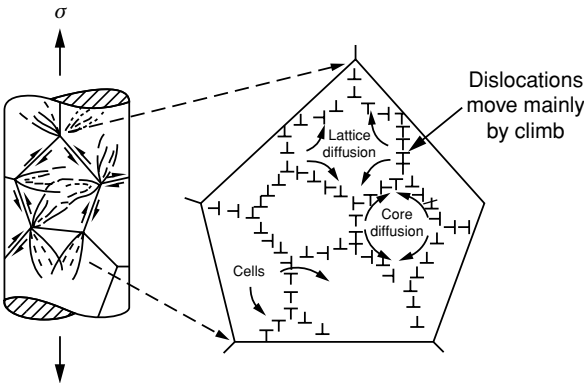


Fig. 2.63. Schematic illustration of the formation of cells by climb during power-law creep. After ref. [71].

in steady-state creep is λ ; if the dislocation density is ρ , it follows to a first approximation that $\lambda = 1/\sqrt{\rho}$. The terms λ and ρ are then state variables which are characteristic of the metallurgical damage induced in the structure. The accumulated dislocation density, ρ , is proportional to the creep strain, ϵ , such that $\dot{\rho} = M\dot{\epsilon}$ [78], where M is a dislocation multiplication constant. To proceed, a law is required for the coarsening of the dislocation cell structure which is responsible for creep softening; consistent with a lattice-diffusion-controlled coarsening, which is analogous to three-dimensional grain growth [79], one expects $\lambda^3 - \lambda_0^3 = KDt$, where t is time, D is a diffusion coefficient, K is a kinetic constant and λ_0 is the initial value of λ . There is some experimental evidence in support of this; it has been shown in ref. [80], for example, that, for low-angle grain boundaries, the activation energy for grain-boundary migration is indistinguishable from that for self-diffusion.

A steady-state creep rate implies a constant sub-grain structure such that ρ and λ are invariant with time; the increments $d\rho_{\text{hardening}}$ and $d\rho_{\text{softening}}$ due to the increments of strain, $d\epsilon$, and sub-grain size, $d\lambda$, are such that

$$d\rho_{\text{total}} = d\rho_{\text{softening}} + d\rho_{\text{hardening}} = 0 \quad (2.43)$$

and hence of equal magnitude but opposite sign. Now, $d\rho_{\text{hardening}}$ is equivalent to $M d\epsilon$ and $d\rho_{\text{softening}}$ is equal to $-2\rho\sqrt{\rho} d\lambda = -[2\rho\sqrt{\rho}KD/(3\lambda^2)] dt$. Inserting these expressions into Equation (2.43), one has

$$\frac{d\epsilon}{dt} = \frac{2\rho\sqrt{\rho}KD}{3\lambda^2M} = \frac{2}{3} \times \frac{1}{\lambda^5} \times \frac{KD}{M} \quad (2.44)$$

so that the strain rate is inversely proportional to the fifth power of the sub-grain size. The stress to give this strain rate, $\dot{\epsilon}$, needs to overcome the obstacles at a spacing of λ ; it is approximately equal to the Orowan stress, Gb/λ . Placing the stress, σ , equal to this estimate, and combining with Equation (2.44), one can show that

$$\frac{d\epsilon}{dt} = \frac{2}{3} \left(\frac{\sigma}{G} \right)^5 \frac{KD}{Mb^5} \quad (2.45)$$

so that one arrives at the fifth power for the stress dependence. Note that the creep strain rate is then proportional to the diffusion coefficient, D , and inversely proportional to the dislocation multiplication constant, M . Writing D in terms of the activation energy, Q , and pre-exponential term, D_0 , and placing $A = 2KD_0/(3Mb^5)$ yields

$$\frac{d\epsilon}{dt} = A \left(\frac{\sigma}{G} \right)^5 \exp \left\{ -\frac{Q}{RT} \right\} \quad (2.46)$$

which has the familiar form of Equation (1.7) from Chapter 1.

In fact, the evidence for pure metals suggests that the constant, A , shows a dependence on the stacking fault energy, γ_{SF} , as evidenced by the proportionality of the normalised secondary creep rate, $\dot{\epsilon}_{\text{ss}}kT/DGb$, and the normalised stacking fault energy, γ_{SF}/Gb , when plotted on logarithmic scales [81]; see Figure 2.64. This indicates that the constant, A , of Equation (2.46) can be written

$$A = A' \left(\frac{\gamma_{\text{SF}}}{Gb} \right)^3 \quad (2.47)$$

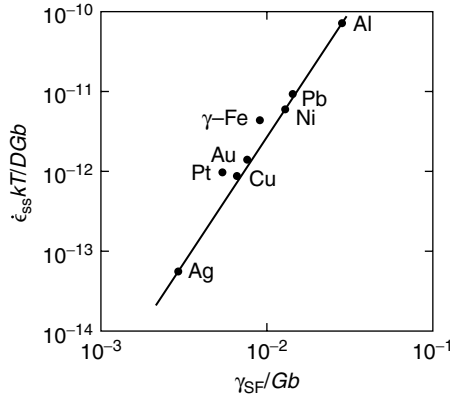


Fig. 2.64. Variation of the function $\dot{\epsilon}_{ss} kT / DGb$ with the normalised stacking fault energy, γ_{SF} / Gb , for various pure metals [81].

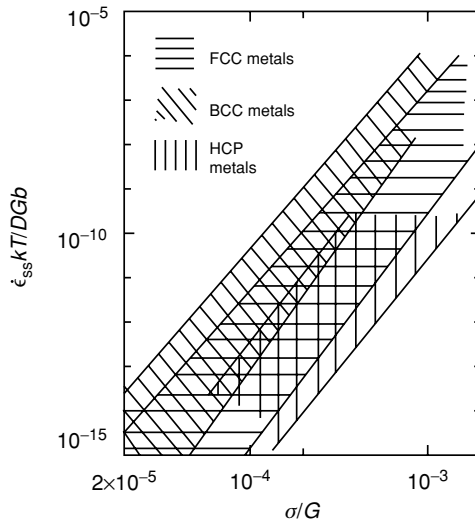


Fig. 2.65. Variation of the function $\dot{\epsilon}_{ss} kT / DGb$ with modulus-compensated stress for FCC, BCC and HCP metals, at a test temperature of 500 °C [81].

so that

$$\frac{d\epsilon}{dt} = A' \left(\frac{\gamma_{SF}}{Gb} \right)^3 \left(\frac{\sigma}{G} \right)^5 \exp \left\{ -\frac{Q}{RT} \right\} \quad (2.48)$$

Although there is some scatter in the data for the pure metals, the steady-state creep rates of those which display the FCC crystal structures are, on average, about one or two orders of magnitude lower than for the BCC case; see Figure 2.65 [72,81]. The HCP metals are even lower still, presumably on account of the limited number of slip systems available. These findings lend further support to the choice of nickel as the solvent for the superalloys.

2.4.2 Creep strengthening in nickel alloys by solid-solution strengthening

Dennison *et al.* [82] have investigated the influence of solute additions on the creep resistance of polycrystalline nickel, of grain size $\sim 80 \mu\text{m}$. Alloying was carried out with gold, palladium, rhodium or iridium at concentrations of 0.1 at% – in each case, either an

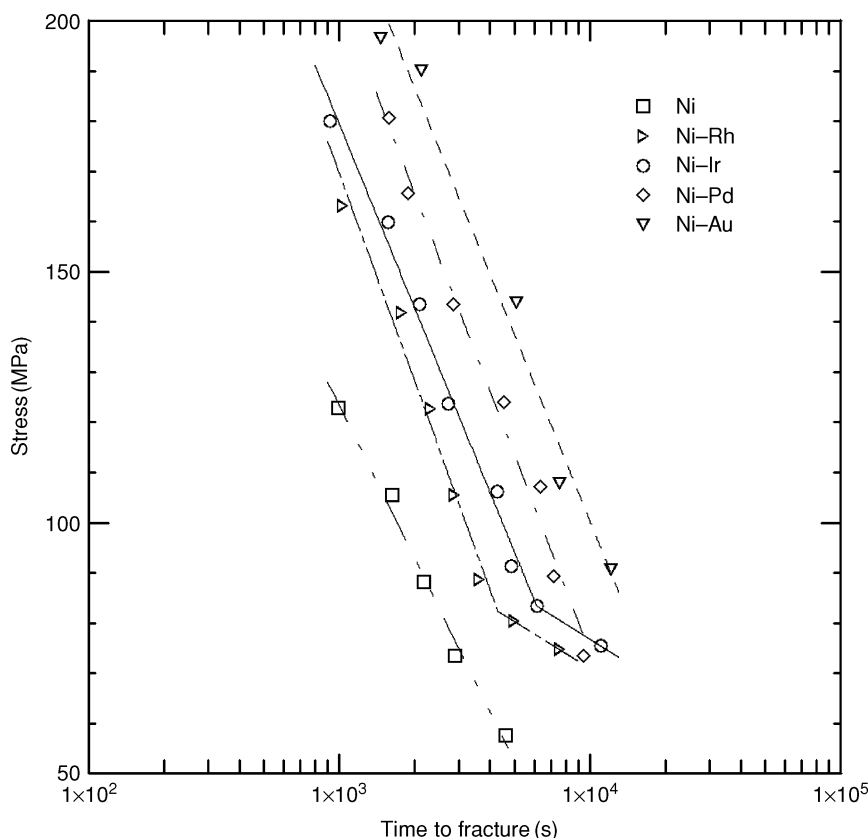


Fig. 2.66. The relationship between applied stress and time to fracture for polycrystalline nickel doped with various elements at a concentration of 0.1 at%, at 500 °C [82].

extensive or complete range of solid solutions is formed with nickel; moreover, the atomic sizes of the solutes are greater than that of the solvent. Testing was carried out at 500 °C and 600 °C under constant load conditions. The solutes were observed to have a profound effect on the creep resistance (see Figure 2.66), with Au and Rh being the most and least potent strengtheners, respectively. The amount of primary creep strain was found to increase in the order rhodium (4%), iridium (5%), palladium (6%) and gold (6%) and compared with nickel at 2%; the values did not depend strongly upon the stress level applied. The data were collated with those from previous work which involved doping with Co, Si and Sn; see Figure 2.67. The results confirm unambiguously that a strong correlation exists between the creep-strengthening increment and the percentage difference in atomic size between the solute and solvent (nickel) atoms. As expected, the degree of creep strengthening increases with the concentration of solute added; see Figure 2.68 [83].

Sherby and Burke [84] introduced a classification for solid-solution-strengthened materials which can be used to interpret the effects produced by alloying. It was realised that a great number of binary alloys behave similarly to pure metals at all concentrations across the phase diagram, but that this is not always so. Alloys whose behaviour is different

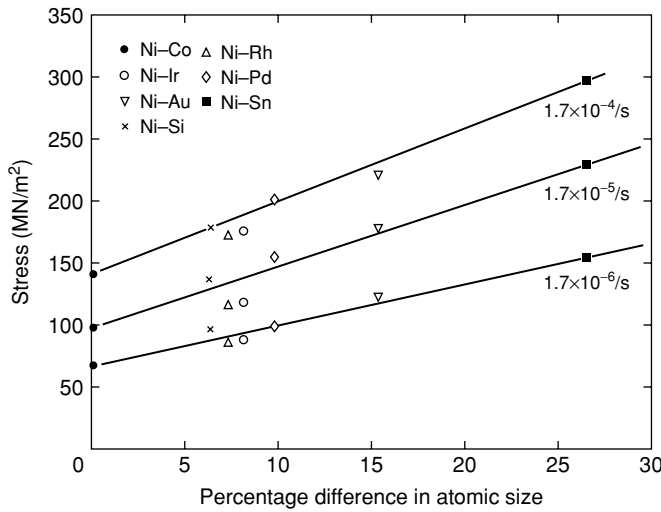


Fig. 2.67. The relationships between the percentage difference in atomic size of solute and solvent atoms and the creep resistance (defined as the stress for a given steady-state creep strain rate) of various dilute nickel alloys at 500 °C [82].

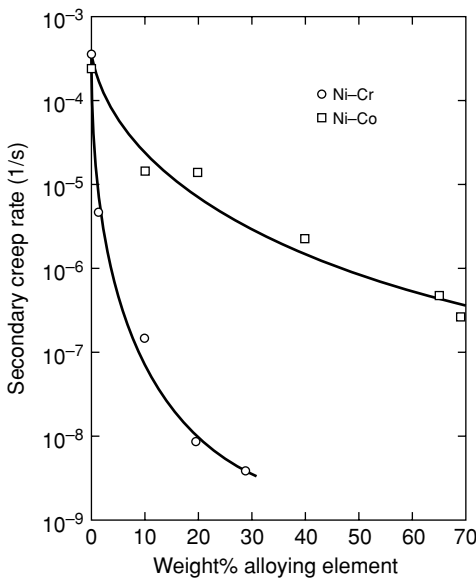


Fig. 2.68. Variation of the secondary creep rate of nickel with cobalt and chromium content during creep at 500 °C. Data taken from ref. [83].

from pure metals were termed Class I alloys and those behaving similarly designated Class II. Thus, Class II alloys exhibit a power-law dependence of secondary creep rate with n equalling five; these alloys are found to show significant primary creep and are influenced by changes in the stacking fault energy. Conversely, Class I alloys were found to exhibit an exponent n of about three; these do not show significant primary creep and are not influenced by the stacking fault energy. Thus, in the Ni–Au binary system, for example, which shows complete solid solubility, the stress exponent changes from five for the pure-end constituents

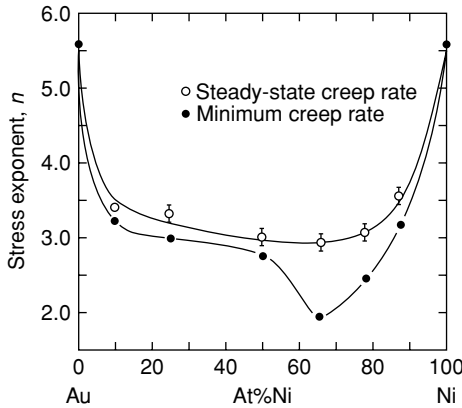


Fig. 2.69. Value of the stress exponent versus alloy content for Ni-Au alloys measured during compression creep; constant stress conditions at a test temperature of 860 °C [85].

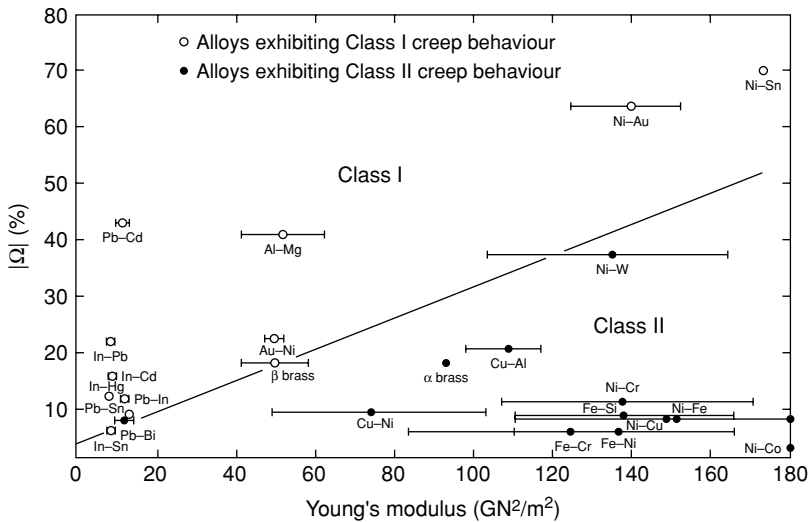


Fig. 2.70. Classification of Class I and Class II alloy behaviour in terms of the misfit in size between solute and solvent atoms, Ω , and the elastic modulus. After ref. [86].

to about three for alloys between the two; see Figure 2.69 [85]. Class I alloys are thought to deform by a glide-controlled process in which interaction between the solute atoms and dislocations leads to a ‘linear viscous’ motion of dislocations. Class I-type behaviour is expected when the atomic misfit between solute and solvent is large, since elastic interactions with dislocations are then favoured. The σ^3 dependence of the linear viscous glide behaviour can be rationalised by noting that the net dislocation velocity, v , is expected to be proportional to the applied stress, σ ; i.e. $v \propto \sigma$. As in the case of pure metals, $\lambda = 1/\sqrt{\rho}$ and $\sigma = Gb/\lambda$, so that the dislocation density, ρ , is expected to be proportional to σ^2 . The third-power dependence then follows from Orowan’s equation, $\dot{\epsilon} = \rho bv$. This is the essence of Weertman’s model of linear viscous creep; see ref. [75].

With these ideas in mind, Cannon and Sherby [86] have noted that Class I behaviour is expected when the size misfit, Ω , is high and the Young’s modulus, E , is low; see Figure 2.70.

This observation can be rationalised as follows. For Class II behaviour – in which the climb rate is expected to be rate-controlling – the creep strain rate, $\dot{\epsilon}$, can be represented by $A''(\sigma/E)^5$. For Class I behaviour – controlled by linear viscous glide – one can take $\dot{\epsilon}$ equal to $K(\sigma/E)^3$ by analogy. Here, A'' and K are constants which depend upon temperature and other factors such as the diffusion coefficient. Treating the two phenomena of glide and climb as sequential processes, one can demonstrate that the net strain rate, $\dot{\epsilon}_{\text{net}}$, is given by

$$\dot{\epsilon}_{\text{net}} = \frac{K A'' \sigma^5}{E^3 (K E^2 + A'' \sigma^2)} \quad (2.49)$$

so that one recovers Class I or Class II behaviour when $K E^2 \ll A'' \sigma^2$ or $K E^2 \gg A'' \sigma^2$, respectively. The dependence upon E follows immediately from this argument; since a large Ω acts to decrease K , all is consistent. Mohamed and Langdon [87] have developed a more sophisticated version of this model which takes the misfit more explicitly into account and includes also the influence of the stacking fault energy, γ_{SF} , in the expression for Class I behaviour.

It is interesting to consider the location on Figure 2.70 of the various alloying elements pertinent to commercial superalloys. The Young's modulus, E , is, in fact, not substantially altered by alloying; hence the data points are clustered around a value of about 130 GPa. The elements taken from the 3d row of transition metals (for example, Fe, Co, Cr) have atomic radii which are not substantially different from that of nickel; consequently the misfit, Ω , is small and therefore Class II behaviour is observed in Ni–Fe, Ni–Co and Ni–Cr binary alloys. The largest misfit is for Ni–Au, which displays Class I behaviour. Ni–W lies in the Class II regime but close to the Class I/Class II boundary. On the basis of the atomic size of their solute, Ni–Mo, Ni–Re and Ni–Ru alloys are expected to lie below Ni–W in the Class II regime, but this remains to be demonstrated.

2.4.3 Creep strengthening in nickel alloys by precipitation hardening

The presence of the γ' phase, which is promoted by alloying with Al, Ti and Ta, has a profound influence on the creep resistance of the nickel alloys. Gibbons and Hopkins [88] have studied the creep behaviour of a series of polycrystalline alloys based upon Nimonic 101 but possessing varying Al and Ti levels and thus volume fractions of the γ' phase. The results (see Figure 2.71, in which the minimum creep rate and life to rupture are plotted against the applied stress) indicate that, provided the stress level is not too high (in which case the γ' particles are sheared), the performance is improved substantially. This strong precipitation-hardening effect was exploited by the International Nickel Company in a series of wrought superalloys used for the turbine blading of early jet engines produced in the 1940s and 1950s; these were strengthened by progressively increasing the fraction of the γ' phase. For example, the Nimonic 80A, 90, 105 and 115 alloys, the compositions of which are given in Table 2.1, have γ' fractions of approximately 0.17, 0.18, 0.32 and 0.62, respectively; when creep resistance is measured with respect to the temperature for 1000 h creep life at 137 MPa, the last of these alloys is better than the first by about 120 °C. The data are plotted on Figure 1.17, which illustrates the evolution of these early alloys.

Gibbons and Hopkins [88] found also that the γ' strengthening effect was most potent when the grain size was large; see Figure 2.72. This effect occurs because the prominent

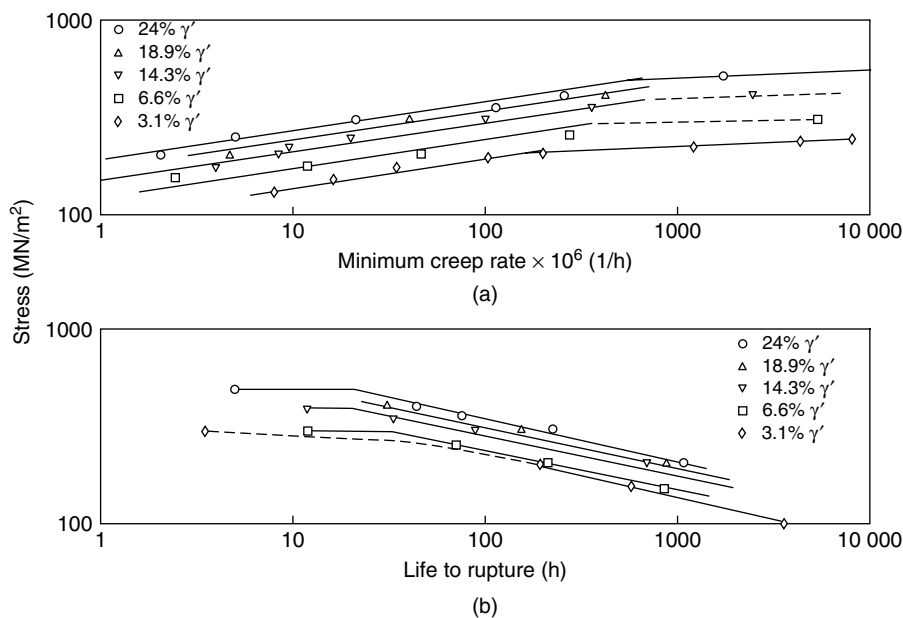


Fig. 2.71. Variation of the (a) minimum creep rate and (b) life to rupture for a number of polycrystalline superalloys with varying fractions of the γ' phase [88].

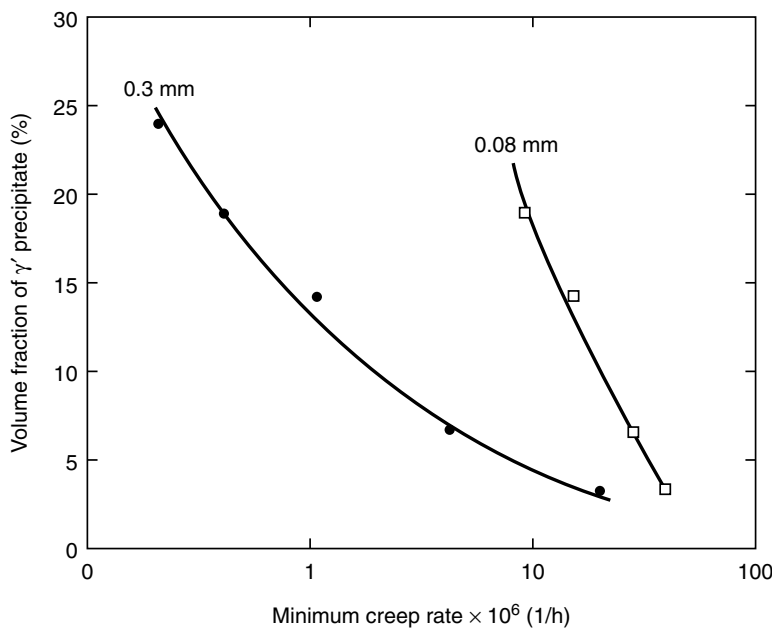


Fig. 2.72. Minimum creep rate of coarse- and fine-grained superalloys as a function of volume fraction of precipitate at 800 °C and 150 MPa [88].

Table 2.1. *The nominal compositions of the Nimonic series of superalloys, in weight%*

Alloy	C	Cr	Ti	Al	Co	Mo	Ni
Nimonic 80A	0.07	19.5	2.4	1.4	—	—	Bal
Nimonic 90	0.08	19.5	2.4	1.4	17	—	Bal
Nimonic 105	0.13	15.0	4.7	1.3	20	5.0	Bal
Nimonic 115	0.15	14.5	3.8	5.0	13	3.3	Bal

form of creep damage in polycrystalline nickel alloys is creep cavitation at the γ -grain boundaries – this becomes less influential as the grain size increases. This effect is exploited in the single-crystal superalloys; a full discussion of these is presented in Chapter 3, but at this point it should be acknowledged that the single-crystal superalloys behave under creep conditions in a very different manner from the polycrystalline alloys from which they were derived. The differences have been summarised succinctly by McLean [89]. First, the shape of the creep curve of the single-crystal superalloys is different; over wide ranges of temperature and applied stress there is no substantial secondary or ‘steady-state’ regime, but instead, over most of the life, the creep rate increases progressively. However, there is no obvious sign of the cavitation or cracking which causes tertiary creep in simple metals. Second, the values of the stress exponent, n , and creep activation energy, Q , derived by fitting the power law of Equation (1.7) to the creep data are typically greater than 8 and about three times the activation energy for self-diffusion of nickel – these findings imply a breakdown of power-law creep so that other theories are required. These are presented in Chapter 3. Third, single-crystal superalloys show a progressive increase in the dislocation density over all of the creep life; there is no evidence that a steady-state is reached, and indeed the dislocation densities are very low compared with simple metals. Finally, pre-straining of single-crystal superalloys weakens them in creep. This is in contrast to the simple metals, which are strengthened if the material is pre-conditioned in this way. This characteristic is, however, not limited to the single-crystal superalloys – the polycrystalline superalloys, such as Nimonic 80A, which possess a significant fraction of γ' behave in this way too [90]; see Figure 2.73. Despite the fact that grain-boundary cavitation is the predominant form of creep damage near the end of life, this confirms that it is the mobile dislocation density which controls the evolution of creep strain at short times, even in polycrystalline superalloys.

2.5 Summary

The physical metallurgy of nickel and its alloys has been reviewed. The high-temperature properties depend crucially upon the combination of alloying elements chosen, since these promote precipitation hardening (most often by the ordered phase γ' , but also by γ'' in iron-rich alloys) and solid-solution strengthening. Additions of Al, Ti and Ta – the important γ' forming elements – are particularly critical if adequate strength levels are to be achieved.

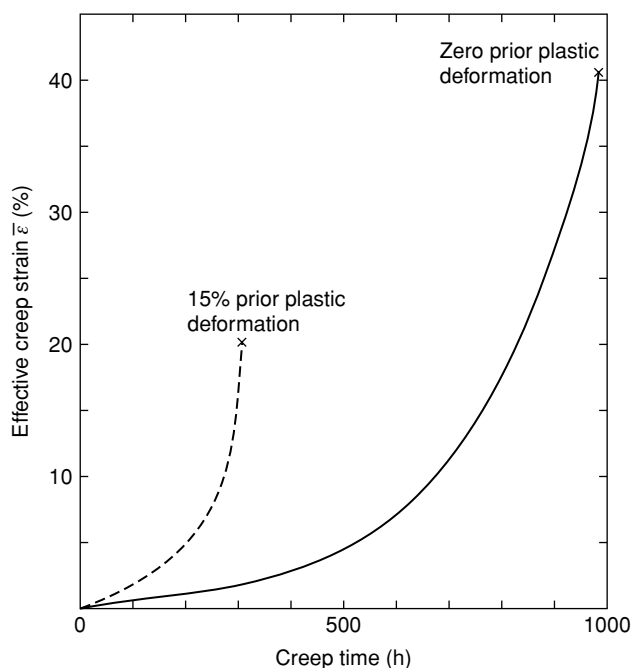


Fig. 2.73. Torsional creep data determined from hollow specimens of the Nimonic 80A polycrystalline superalloy, tested at 750 °C and an effective stress of 234 MPa [90]. The dashed line corresponds to a specimen which had been pre-strained plastically to 15% prior to creep testing. Note that creep softening is induced by the prior deformation.

The γ matrix phase possesses a cube–cube orientation relationship with the γ' precipitate phase which forms from it; these display the face-centred cubic (FCC) and $L1_2$ crystal structures, respectively. Provided that the elemental partitioning between γ and γ' is optimal such that the γ and γ' lattice parameters are balanced, the lattice misfit is small, the γ/γ' interfaces are coherent and the elastic strain energy is minimised. Other phases of importance are the carbides MC , $M_{23}C_6$ and M_6C , which are rich in Cr, Hf, Mo, Ti, Ta and W, and the topologically close-packed (TCP) phases μ , P , σ and R , which contain substantial concentrations of Cr, Mo and Re.

Defects are present – despite the crystalline nature of the phases – which control the mechanical properties exhibited. These are of various types: point defects such as vacancies; line defects such as dislocations (usually in dissociated form); and planar defects such as stacking faults. The stacking fault energy of nickel is one of the largest of all FCC metals at 250 mJ/m², so that partial dislocations in unalloyed γ are closely spaced. Dissociation is promoted by alloying additions, which decrease the stacking fault energy such that creep processes are hindered. The activation energy for solute–vacancy exchange depends strongly upon the solute species, with Re possessing a value (in excess of 160 kJ/mol) larger than for all others in the d block of transition metals; thus, Re is expected to retard diffusion-controlled processes in nickel alloys. Owing to the ordered structure, defects in γ' are complex. The anti-phase boundary (APB) is one type of planar defect which separates